GATE 2027
Robotics and Automation (RA)

Complete topic-wise study notes

Part A: Engineering Mathematics · Mechatronics · Robotics & Automation
Part B1: Electrical Engineering · Part B2: Mechanical Engineering

Organized exactly by the official GATE 2027 RA syllabus (IIT Madras).
Concise exam notes: definitions, key results, formulas and standard cases.

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Part A.1 — Engineering Mathematics

1. Linear Algebra

Matrix basics

- **Transpose:** $(A^T)_{ij} = A_{ji}$; $(AB)^T = B^TA^T$; $(A^{-1})^T = (A^T)^{-1}$.
- **Symmetric:** $A^T = A$. **Skew-symmetric:** $A^T = -A$ (diagonal = 0; odd order ⇒ det = 0). Any $A = \frac{1}{2}(A+A^T) + \frac{1}{2}(A-A^T)$.
- **Orthogonal:** $A^TA = I \Rightarrow A^{-1} = A^T$, det = ±1, columns orthonormal. Rotation matrices are orthogonal with det = +1.
- **Trace:** sum of diagonal = sum of eigenvalues; tr(AB) = tr(BA).
- **Adjoint:** adj(A) = transpose of cofactor matrix. $A\cdot adj(A) = det(A)\cdot I \Rightarrow A^{-1} = adj(A)/det(A)$. $det(adj\ A) = (det\ A)^{n-1}$ for n×n. $adj(adj\ A) = (det\ A)^{n-2}\cdot A$.

Determinant

- $det(AB) = det\ A \cdot det\ B$; $det(A^T) = det\ A$; $det(kA) = k^n\ det\ A$; $det(A^{-1}) = 1/det\ A$.
- Swapping two rows changes sign; adding a multiple of one row to another leaves det unchanged; two equal/proportional rows ⇒ det = 0.
- Triangular matrix: det = product of diagonal. det = product of eigenvalues.

Rank

- Rank = number of linearly independent rows (= columns) = number of non-zero rows in row-echelon form = order of largest non-zero minor.
- $rank(AB) \le \min(rank\ A, rank\ B)$. $rank(A) = rank(A^T)$. Rank–nullity: rank + nullity = n (number of columns).

Systems of linear equations Ax = b (n unknowns)

Condition	Result
$rank(A) \neq rank([A\|b])$	Inconsistent — no solution
$rank(A) = rank([A\|b]) = n$	Unique solution
$rank(A) = rank([A\|b]) = r < n$	Infinitely many solutions; n − r free parameters

- Homogeneous Ax = 0 is always consistent; non-trivial solution exists iff rank < n (for square A, iff det A = 0).
- Methods: Gaussian elimination, Gauss–Jordan, Cramer's rule ($x_i = det\ A_i / det\ A$), LU decomposition.

Eigenvalues and eigenvectors

- $Ax = \lambda x$, $x \neq 0$. Characteristic equation: $det(A - \lambda I) = 0$.
- $\Sigma\lambda_i$ = trace, $\Pi\lambda_i$ = det. Eigenvalues of triangular/diagonal matrix = diagonal entries.
- Eigenvalues of A^{-1}: $1/\lambda$; A^k: λ^k; A + kI: λ + k; kA: kλ; A^T: same λ. Eigenvectors stay the same (except for A^T).
- Real symmetric ⇒ real eigenvalues, orthogonal eigenvectors. Skew-symmetric ⇒ zero or purely imaginary. Orthogonal ⇒ $|\lambda| = 1$.
- Distinct eigenvalues ⇒ linearly independent eigenvectors ⇒ diagonalizable: $A = PDP^{-1}$.
- **Cayley–Hamilton:** every matrix satisfies its own characteristic equation (useful to find A^{-1} and powers).

- Algebraic multiplicity ≥ geometric multiplicity (= n − rank(A − λI)).

2. Calculus

Limits, continuity, differentiability

- Limit exists iff LHL = RHL. Continuous at a iff $\lim_{x\to a} f(x) = f(a)$.
- Differentiable ⇒ continuous (not conversely; e.g. |x| at 0).
- Standard limits: $\sin x/x \to 1$; $(1 + x)^{1/x} \to e$; $(e^x - 1)/x \to 1$; $(a^x - 1)/x \to \ln a$; $(1 + 1/x)^x \to e$ as $x\to\infty$.

Mean value theorems

- **Rolle:** f continuous on [a,b], differentiable on (a,b), f(a) = f(b) ⇒ ∃ c: f′(c) = 0.
- **Lagrange (MVT):** ∃ c ∈ (a,b): f′(c) = [f(b) − f(a)]/(b − a).
- **Cauchy:** [f(b) − f(a)]/[g(b) − g(a)] = f′(c)/g′(c).

Indeterminate forms

- 0/0, ∞/∞ → L'Hôpital: lim f/g = lim f′/g′.
- 0·∞ → rewrite as 0/0. ∞ − ∞ → combine into one fraction. 1^∞, 0^0, ∞^0 → take log: $L = e^{\lim g\cdot\ln f}$. For 1^∞: $L = e^{\lim g(f-1)}$.

Integrals

- $\int_0^a f(x)dx = \int_0^a f(a-x)dx$. $\int_{-a}^{a} f\,dx = 2\int_0^a f$ (even) or 0 (odd).
- Wallis: $\int_0^{\pi/2} \sin^n x\,dx$ = [(n−1)(n−3)…/n(n−2)…] × (π/2 if n even, 1 if n odd).
- **Improper integrals:** infinite limits or unbounded integrand; evaluate as a limit. $\int_1^\infty 1/x^p\,dx$ converges iff p > 1; $\int_0^1 1/x^p\,dx$ converges iff p < 1.
- Gamma: $\Gamma(n) = \int_0^\infty e^{-x}x^{n-1}dx$, $\Gamma(n+1) = n\Gamma(n) = n!$, $\Gamma(½) = \sqrt{\pi}$. $\int_0^\infty e^{-x^2}dx = \sqrt{\pi}/2$.

Double and triple integrals

- Change order of integration by redrawing the region. Polar: dx dy = r dr dθ. Cylindrical: r dr dθ dz. Spherical: $r^2 \sin\varphi\, dr\, d\theta\, d\varphi$.
- Area = ∬ dx dy; volume = ∭ dV; general change of variables uses the Jacobian $|\partial(x,y)/\partial(u,v)|$.

Partial and total derivatives

- Total differential: $du = (\partial u/\partial x)dx + (\partial u/\partial y)dy$. Chain rule: $du/dt = u_x\cdot dx/dt + u_y\cdot dy/dt$.
- **Euler's theorem** (u homogeneous of degree n): $x\,u_x + y\,u_y = n\,u$; second order: $x^2u_{xx} + 2xy\,u_{xy} + y^2u_{yy} = n(n-1)u$.
- Implicit: $dy/dx = -f_x/f_y$.

Taylor series

- One variable: $f(a+h) = f(a) + h\,f'(a) + h^2/2!\,f''(a) + \ldots$ Maclaurin at a = 0.
- Key series: $e^x = \Sigma x^n/n!$; $\sin x = x - x^3/3! + x^5/5! - \ldots$; $\cos x = 1 - x^2/2! + x^4/4! - \ldots$; $\ln(1+x) = x - x^2/2 + x^3/3 - \ldots$; $1/(1-x) = 1 + x + x^2 + \ldots$ (|x| < 1).
- Two variables: $f(a+h, b+k) = f + (h\,f_x + k\,f_y) + (1/2!)(h^2\,f_{xx} + 2hk\,f_{xy} + k^2\,f_{yy}) + \ldots$

Maxima and minima

- One variable: f′ = 0; f″ < 0 ⇒ max, f″ > 0 ⇒ min, f″ = 0 ⇒ check higher derivatives.
- Two variables: $f_x = f_y = 0$. $r = f_{xx}$, $s = f_{xy}$, $t = f_{yy}$. $rt - s^2 > 0$ and r < 0 ⇒ max; $rt - s^2 > 0$ and r > 0 ⇒ min; $rt - s^2 < 0$ ⇒ saddle; = 0 ⇒ inconclusive.
- Constrained: Lagrange multipliers, ∇f = λ∇g.

Vector calculus

- **Gradient** $\nabla\varphi$: vector normal to level surface; directional derivative = $\nabla\varphi\cdot\hat{a}$; maximum rate = $|\nabla\varphi|$.
- **Divergence** $\nabla\cdot F = \partial F_1/\partial x + \partial F_2/\partial y + \partial F_3/\partial z$ (scalar). $\nabla\cdot F = 0 \Rightarrow$ solenoidal.
- **Curl** $\nabla\times F$ (vector). $\nabla\times F = 0 \Rightarrow$ irrotational (conservative, $F = \nabla\varphi$).
- Identities: $\nabla\times(\nabla\varphi) = 0$; $\nabla\cdot(\nabla\times F) = 0$; $\nabla\cdot(\varphi F) = \varphi\nabla\cdot F + F\cdot\nabla\varphi$; $\nabla\times(\varphi F) = \varphi\nabla\times F + \nabla\varphi\times F$; $\nabla\times(\nabla\times F) = \nabla(\nabla\cdot F) - \nabla^2 F$; $\nabla\cdot r = 3$, $\nabla\times r = 0$, $\nabla r^n = n\, r^{n-2}\, r$.
- **Green:** $\oint(M\,dx + N\,dy) = \iint(\partial N/\partial x - \partial M/\partial y)\,dA$. **Stokes:** $\oint F\cdot dr = \iint(\nabla\times F)\cdot n\,dS$. **Gauss divergence:** $\oiint F\cdot n\,dS = \iiint \nabla\cdot F\,dV$.

3. Differential Equations

First order

- **Variable separable:** $f(x)dx = g(y)dy$.
- **Homogeneous** $dy/dx = f(y/x)$: put $y = vx$.
- **Exact:** $M\,dx + N\,dy = 0$ exact iff $\partial M/\partial y = \partial N/\partial x$. Solution: $\int M\,dx$ (y const) + $\int$(terms of N without x) $dy = c$. Integrating factors: $(M_y - N_x)/N = f(x) \Rightarrow IF = e^{\int f\,dx}$.
- **Linear:** $y' + P(x)y = Q(x)$. $IF = e^{\int P\,dx}$; $y\cdot IF = \int Q\cdot IF\,dx + c$.
- **Bernoulli** (nonlinear): $y' + Py = Qy^n$; put $z = y^{1-n}$ to make it linear.

Higher order linear with constant coefficients

- Solution y = C.F. + P.I. Auxiliary equation from $D \to m$.
- C.F.: distinct real roots → $\Sigma c_i e^{m_i x}$; repeated root m → $(c_1 + c_2 x)e^{mx}$; complex $\alpha \pm i\beta$ → $e^{\alpha x}(c_1\cos\beta x + c_2\sin\beta x)$.
- P.I.: $e^{ax} \to e^{ax}/f(a)$ (if $f(a) = 0$, multiply by x and differentiate f). sin ax, cos ax → replace D^2 by $-a^2$. x^n → expand $[f(D)]^{-1}$ as a power series. $e^{ax}V \to e^{ax}\cdot[1/f(D+a)]V$.

Euler–Cauchy equation

- $x^2y'' + a\,x\,y' + b\,y = R(x)$. Substitute $x = e^t$ ($xD = \theta$, $x^2D^2 = \theta(\theta-1)$) → constant-coefficient ODE in t.
- Equivalently try $y = x^m$: $m(m-1) + am + b = 0$. Repeated root m ⇒ $y = (c_1 + c_2 \ln x)x^m$.

IVP and BVP

- IVP: conditions at one point ($y(x_0)$, $y'(x_0)$). BVP: conditions at two points; may have no or infinitely many solutions.

Laplace transforms

f(t)	F(s)	f(t)	F(s)
1	$1/s$	t^n	$n!/s^{n+1}$
e^{at}	$1/(s-a)$	sin at	$a/(s^2+a^2)$
cos at	$s/(s^2+a^2)$	sinh at	$a/(s^2-a^2)$
$\delta(t)$	1	u(t−a)	e^{-as}/s

- Shifting: $L\{e^{at}f\} = F(s-a)$; $L\{f(t-a)u(t-a)\} = e^{-as}F(s)$.
- Derivative: $L\{f'\} = sF - f(0)$; $L\{f''\} = s^2F - s\,f(0) - f'(0)$. Integral: $L\{\int_0^t f\} = F/s$.
- $L\{t\,f\} = -F'(s)$; $L\{f/t\} = \int_s^\infty F\,ds$. Convolution: $L\{f*g\} = F\cdot G$.
- Initial value: $f(0^+) = \lim_{s\to\infty} sF(s)$. Final value: $f(\infty) = \lim_{s\to 0} sF(s)$ (if poles of sF in LHP).

PDEs: heat, wave, Laplace

- Classification of $A\,u_{xx} + B\,u_{xy} + C\,u_{yy}$: $B^2 - 4AC < 0$ elliptic (Laplace), $= 0$ parabolic (heat), > 0 hyperbolic (wave).
- **Heat:** $u_t = c^2 u_{xx}$; with $u(0,t) = u(L,t) = 0$: $u = \Sigma\, b_n \sin(n\pi x/L)\, e^{-c^2 n^2 \pi^2 t/L^2}$.
- **Wave:** $u_{tt} = c^2 u_{xx}$; fixed ends: $u = \Sigma\,[A_n \cos(n\pi ct/L) + B_n \sin(n\pi ct/L)] \sin(n\pi x/L)$. D'Alembert: $u = \tfrac{1}{2}[f(x+ct) + f(x-ct)] + (1/2c)\int g$.
- **Laplace:** $u_{xx} + u_{yy} = 0$ (steady-state heat). Method: separation of variables $u = X(x)T(t)$.

Fourier series

- Period $2L$: $f(x) = a_0/2 + \Sigma[a_n \cos(n\pi x/L) + b_n \sin(n\pi x/L)]$; $a_n = (1/L)\int_{-l}^{L} f \cos(n\pi x/L)\,dx$, $b_n = (1/L)\int_{-l}^{L} f \sin(n\pi x/L)\,dx$.
- Even $f \Rightarrow$ cosine series ($b_n = 0$). Odd $f \Rightarrow$ sine series ($a_n = 0$).
- At a jump discontinuity the series converges to the average of left and right limits (Dirichlet).
- Parseval: $(1/L)\int f^2 = a_0^2/2 + \Sigma(a_n^2 + b_n^2)$.

4. Probability and Statistics

- **Axioms:** $P(A) \geq 0$; $P(S) = 1$; for disjoint events $P(A \cup B) = P(A) + P(B)$.
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. Independent $\Rightarrow P(A \cap B) = P(A)P(B)$.
- **Conditional:** $P(A|B) = P(A \cap B)/P(B)$. **Total probability:** $P(B) = \Sigma P(B|A_i)P(A_i)$. **Bayes:** $P(A_i|B) = P(B|A_i)P(A_i)/\Sigma P(B|A_j)P(A_j)$.
- **Central tendency:** mean $= \Sigma x/n$; median = middle value; mode = most frequent. Empirical: mode ≈ 3 median $-$ 2 mean.
- **Dispersion:** variance $\sigma^2 = E[X^2] - (E[X])^2$; SD $= \sigma$. $\mathrm{Var}(aX + b) = a^2 \mathrm{Var}(X)$.
- **Random variable:** discrete (PMF) or continuous (PDF). **PDF** $f(x) \geq 0$, $\int f = 1$, $P(a<X<b) = \int_a^b f$. **CDF** $F(x) = P(X \leq x)$, non-decreasing, $F(-\infty) = 0$, $F(\infty) = 1$, $f = F'$.
- $E[X] = \Sigma x\, p(x)$ or $\int x\, f(x)\,dx$; $E[aX + bY] = aE[X] + bE[Y]$.

Distribution	PMF / PDF	Mean	Variance
Binomial (n, p)	${}^nC_x\, p^x\, q^{n-x}$	np	npq
Poisson (λ)	$e^{-\lambda}\lambda^x/x!$	λ	λ
Uniform (a, b)	$1/(b-a)$	$(a+b)/2$	$(b-a)^2/12$
Exponential (λ)	$\lambda e^{-\lambda x}$	$1/\lambda$	$1/\lambda^2$
Normal (μ, σ²)	$(1/\sigma\sqrt{2\pi})\, e^{-(x-\mu)^2/2\sigma^2}$	μ	σ^2

- Poisson approximates binomial when n large, p small, $\lambda = np$. Normal: $z = (x - \mu)/\sigma$; 68–95–99.7 rule for $\pm1\sigma$, $\pm2\sigma$, $\pm3\sigma$.
- **Hypothesis testing:** H_0 (null) vs H_1. Type I error $\alpha = P(\text{reject } H_0 \mid H_0 \text{ true})$; Type II $\beta = P(\text{accept } H_0 \mid H_0 \text{ false})$; power $= 1 - \beta$.
- z-test (σ known, large n): $z = (\bar{x} - \mu)/(\sigma/\sqrt{n})$. t-test (σ unknown, small n): $t = (\bar{x} - \mu)/(s/\sqrt{n})$, $n-1$ dof. χ^2 test for goodness of fit/variance. Reject H_0 if |statistic| > critical value (or p-value $< \alpha$).

5. Numerical Methods

Algebraic and transcendental equations

- **Bisection:** $f(a)f(b) < 0$, halve the interval; linear convergence; error after n steps $\leq (b-a)/2^n$.

- **Regula falsi:** $x = (a\,f(b) - b\,f(a))/(f(b) - f(a))$.
- **Newton–Raphson:** $x_{n+1} = x_n - f(x_n)/f'(x_n)$; quadratic convergence (order 2). Square root of N: $x_{n+1} = \tfrac{1}{2}(x_n + N/x_n)$.
- **Secant:** order ≈ 1.618. **Fixed-point** $x = g(x)$ converges if $|g'(x)| < 1$.
- Linear systems: Gauss elimination (direct); Jacobi and Gauss–Seidel (iterative, converge if diagonally dominant; Gauss–Seidel faster).

Numerical integration (h = (b−a)/n)

Rule	Formula	Error order
Trapezoidal	$(h/2)[y_0 + 2(y_1+\ldots+y_{n-1}) + y_n]$	$O(h^2)$ global; exact for linear
Simpson 1/3 (n even)	$(h/3)[y_0 + 4(\text{odd}) + 2(\text{even}) + y_n]$	$O(h^4)$; exact up to cubic
Simpson 3/8 (n multiple of 3)	$(3h/8)[y_0 + 3y_1 + 3y_2 + 2y_3 + \ldots + y_n]$	$O(h^4)$

Finite differences

- Forward $\Delta y = y_1 - y_0$; backward $\nabla y = y_1 - y_0$ (at y_1); central δ. Shift $E = 1 + \Delta$.
- $f'(x) \approx [f(x+h) - f(x)]/h$ $(O(h))$; central $[f(x+h) - f(x-h)]/2h$ $(O(h^2))$; $f'' \approx [f(x+h) - 2f(x) + f(x-h)]/h^2$.

ODE solvers for y′ = f(x, y)

- **Explicit Euler:** $y_{n+1} = y_n + h\,f(x_n, y_n)$ — first order, local error $O(h^2)$, global $O(h)$.
- **Modified Euler / Heun:** predictor-corrector, second order.
- **RK4:** $k_1 = hf(x,y)$; $k_2 = hf(x+h/2, y+k_1/2)$; $k_3 = hf(x+h/2, y+k_2/2)$; $k_4 = hf(x+h, y+k_3)$; $y_{n+1} = y_n + (k_1 + 2k_2 + 2k_3 + k_4)/6$. Fourth order.
- **Multi-step:** Adams–Bashforth (explicit), Adams–Moulton (implicit), Milne's predictor-corrector — use several previous points.
- Stability: explicit Euler for $y' = \lambda y$ stable if $|1 + h\lambda| \le 1$.

Part A.2 — Basics of Mechatronics

1. Network Elements and Analysis

Elements and sources

- **Ideal voltage source:** fixed V, zero internal resistance (practical: series R). **Ideal current source:** fixed I, infinite internal resistance (practical: parallel R).
- **Dependent sources:** VCVS, VCCS, CCVS, CCCS. Do not switch them off when finding R_{th}; use a test source instead.
- **R:** $v = iR$, $P = i^2R$. **L:** $v = L\,di/dt$, energy $\frac{1}{2}Li^2$, current cannot change instantly, acts as a short at DC. **C:** $i = C\,dv/dt$, energy $\frac{1}{2}Cv^2$, voltage cannot change instantly, acts as an open at DC.
- Series R add; parallel 1/R add. Inductors combine like R; capacitors combine the opposite way. **M (mutual inductance):** $v_2 = M\,di_1/dt$; $M = k\sqrt{(L_1L_2)}$; series aiding $L = L_1 + L_2 + 2M$, opposing $L_1 + L_2 - 2M$.
- Voltage divider: $V_x = V \cdot R_x/\Sigma R$. Current divider: $I_1 = I \cdot R_2/(R_1 + R_2)$.
- Star–delta: $R_Y = R_{ab}R_{ac}/\Sigma R_\Delta$; balanced $R_Y = R_\Delta/3$.

Kirchhoff's laws and solution methods

- **KCL:** Σ currents into a node = 0 (charge conservation). **KVL:** Σ voltages around a loop = 0 (energy conservation).
- **Nodal analysis:** pick a reference node, write KCL at the other (n − 1) nodes in terms of node voltages. A voltage source between two non-reference nodes forms a supernode.
- **Mesh analysis:** write KVL around b − n + 1 meshes (planar circuits). A current source shared by two meshes forms a supermesh.

Network theorems

- **Thevenin:** any linear two-terminal network = V_{th} (open-circuit voltage) in series with R_{th} (sources off: V → short, I → open).
- **Norton:** I_N (short-circuit current) in parallel with $R_N = R_{th}$. $V_{th} = I_N R_{th}$.
- **Superposition:** response = sum of responses to each independent source acting alone (linear circuits only; not valid for power).
- **Maximum power transfer:** DC: $R_L = R_{th}$, $P_{max} = V_{th}^2/4R_{th}$, efficiency 50%. AC: $Z_L = Z_{th}^*$ (conjugate); if only $|Z_L|$ can vary, $R_L = |Z_{th}|$.
- Also know: reciprocity, Tellegen ($\Sigma v_i i_i = 0$), Millman.

Transient response

- First order: $x(t) = x(\infty) + [x(0^+) - x(\infty)]e^{-t/\tau}$. τ = RC or L/R (R = Thevenin resistance seen by C or L).
- After 5τ the response is >99% settled. RC charging: $v_C = V(1 - e^{-t/RC})$.
- Second-order series RLC: $s^2 + (R/L)s + 1/LC = 0$. $\omega_0 = 1/\sqrt{LC}$, $\zeta = (R/2)\sqrt{(C/L)}$. ζ > 1 overdamped, = 1 critically damped, < 1 underdamped (oscillatory with $\omega_d = \omega_0\sqrt{(1-\zeta^2)}$).
- Parallel RLC: $\zeta = (1/2R)\sqrt{(L/C)}$.

Sinusoidal steady state

- Phasor: $v = V_m\cos(\omega t + \varphi) \rightarrow V\angle\varphi$. Impedances: $Z_R = R$, $Z_L = j\omega L$, $Z_C = 1/j\omega C$.

- RMS of sinusoid = $V_m/\sqrt{2}$. Current lags voltage in an inductive circuit, leads in a capacitive circuit.
- **Complex power:** $S = V I^* = P + jQ$. $P = V_{rms}I_{rms}\cos\varphi$ (W), $Q = V_{rms}I_{rms}\sin\varphi$ (VAR), $|S|$ (VA). Power factor = $\cos\varphi = P/|S|$. PF correction adds a shunt capacitor: $Q_C = P(\tan\varphi_1 - \tan\varphi_2)$.

Resonance

- **Series RLC:** $\omega_0 = 1/\sqrt{LC}$; $Z = R$ (minimum), current maximum, PF = 1. $Q = \omega_0 L/R = (1/R)\sqrt{(L/C)}$. Bandwidth = $R/L = \omega_0/Q$. Voltage across L or C = Q × supply voltage.
- **Parallel RLC:** impedance maximum, current minimum. $Q = R\sqrt{(C/L)}$. $BW = 1/RC$.
- Half-power frequencies: $\omega_1\omega_2 = \omega_0^2$.

Two-port networks

Parameters	Equations	Reciprocal if	Symmetric if
Z (open circuit)	$V_1 = Z_{11}I_1 + Z_{12}I_2$; $V_2 = Z_{21}I_1 + Z_{22}I_2$	$Z_{12} = Z_{21}$	$Z_{11} = Z_{22}$
Y (short circuit)	$I_1 = Y_{11}V_1 + Y_{12}V_2$; $I_2 = Y_{21}V_1 + Y_{22}V_2$	$Y_{12} = Y_{21}$	$Y_{11} = Y_{22}$
h (hybrid)	$V_1 = h_{11}I_1 + h_{12}V_2$; $I_2 = h_{21}I_1 + h_{22}V_2$	$h_{12} = -h_{21}$	$\Delta h = 1$
ABCD (transmission)	$V_1 = AV_2 - BI_2$; $I_1 = CV_2 - DI_2$	$AD - BC = 1$	$A = D$

- Cascade connection: multiply ABCD matrices. Series: add Z. Parallel: add Y.

Balanced three-phase circuits

- Star (Y): $V_L = \sqrt{3}\,V_{ph}$ (leads by 30°), $I_L = I_{ph}$. Delta (Δ): $V_L = V_{ph}$, $I_L = \sqrt{3}\,I_{ph}$ (lags by 30°).
- Total power $P = \sqrt{3}\,V_L I_L\cos\varphi = 3V_{ph}I_{ph}\cos\varphi$. $Q = \sqrt{3}\,V_L I_L\sin\varphi$.
- **Two-wattmeter method:** $P = W_1 + W_2$; $\tan\varphi = \sqrt{3}(W_1 - W_2)/(W_1 + W_2)$. At PF 0.5 one wattmeter reads zero; below 0.5 one reads negative.
- Neutral current in a balanced star = 0.

2. Digital Circuits

Boolean algebra

- Laws: identity ($A + 0 = A$, $A\cdot 1 = A$), null ($A + 1 = 1$, $A\cdot 0 = 0$), idempotent, complement ($A + A' = 1$), absorption ($A + AB = A$, $A(A + B) = A$), $A + A'B = A + B$.
- **De Morgan:** $(A + B)' = A'B'$; $(AB)' = A' + B'$. **Consensus:** $AB + A'C + BC = AB + A'C$.
- SOP (minterms Σm) and POS (maxterms ΠM). K-maps: group 1s in powers of 2 (adjacent wraps around); don't-cares can be used for larger groups.
- NAND and NOR are universal gates. XOR: $A \oplus B = A'B + AB'$; XNOR is equality. $A \oplus A = 0$, $A \oplus 1 = A'$.
- Number systems: binary, octal, hex, BCD, Gray ($g_i = b_i \oplus b_{i+1}$). 2's complement: invert and add 1; n-bit range -2^{n-1} to $2^{n-1} - 1$.

Combinational circuits

- **Half adder:** $S = A \oplus B$, $C = AB$. **Full adder:** $S = A \oplus B \oplus C_{in}$, $C = AB + C_{in}(A \oplus B)$.
- **Multiplexer (2^n:1):** n select lines choose one of 2^n inputs. A 2^n:1 MUX can implement any n-variable function (or n+1 variables using the last variable at data inputs).
- **Demultiplexer:** one input to one of 2^n outputs.
- **Decoder ($n \to 2^n$):** activates one output per input code; decoder + OR gate implements any SOP.
- **Encoder ($2^n \to n$):** inverse; **priority encoder** outputs the highest-priority active input.

- Comparators, code converters, parity generators.

Sequential circuits

Flip-flop	Characteristic equation	Notes
SR	$Q^+ = S + R'Q$ (SR = 0)	S = R = 1 forbidden
JK	$Q^+ = JQ' + K'Q$	J = K = 1 toggles; race-around in level-triggered → use master-slave/edge
D	$Q^+ = D$	Delay/data latch
T	$Q^+ = T \oplus Q$	Toggles when T = 1

- Latches are level-triggered; flip-flops are edge-triggered. Setup and hold time must be met.
- **Counters:** asynchronous (ripple): clock passes through FFs, delay = $n \cdot t_{pd}$. Synchronous: common clock, faster. n flip-flops → mod up to 2^n.
- **Ring counter:** n FFs → mod n. **Johnson (twisted ring):** n FFs → mod 2n.
- Shift registers: SISO, SIPO, PISO, PIPO. Mealy output depends on state and input; Moore depends on state only.
- Max clock frequency $f_{max} = 1/(t_{pd,FF} + t_{comb} + t_{setup})$.

3. Sensors and Signal Conditioning

- **Static characteristics:** accuracy, precision, resolution, sensitivity (Δoutput/Δinput), linearity, hysteresis, range/span, drift, dead zone. **Dynamic:** response time, time constant, bandwidth.
- **Resistive:** potentiometer (displacement, $V_{out} = V \cdot x/L$; loading error when meter R is finite), strain gauge (gauge factor $GF = (\Delta R/R)/\varepsilon \approx 2$ for metals), RTD ($R = R_0(1 + \alpha T)$, platinum Pt100), thermistor (NTC, nonlinear, $R = R_0 e^{\beta(1/T - 1/T_0)}$), LDR.
- **Capacitive:** $C = \varepsilon A/d$. Vary d (nonlinear, high sensitivity), A (linear) or ε (level sensing). Differential arrangement improves linearity.
- **Inductive:** variable reluctance, **LVDT** (primary + two secondaries in series opposition; output ∝ displacement, phase gives direction; null at centre; needs phase-sensitive demodulation). RVDT for angle.
- **Piezoelectric:** charge $q = d \cdot F$ (d = charge sensitivity, C/N). Only for dynamic (not static) measurement because charge leaks. Needs a charge amplifier. Used in accelerometers, force/pressure sensors.
- **Hall effect:** $V_H = BI/(n\,q\,t)$. Used for proximity, current, speed sensing, BLDC commutation.
- **Thermocouple:** Seebeck effect, needs cold-junction compensation.

Signal conditioning

- **Wheatstone bridge:** balanced when $R_1R_4 = R_2R_3$. Quarter bridge (1 active gauge): $V_o \approx V \cdot GF \cdot \varepsilon/4$; half bridge ×2; full bridge (4 active): $V \cdot GF \cdot \varepsilon$. Dummy gauge for temperature compensation.
- Instrumentation amplifier (high CMRR, high input impedance), filtering, linearization, isolation, ADC, excitation, cold-junction compensation, demodulation.

Transducers for industrial instrumentation

Quantity	Transducers
Linear displacement	Potentiometer, LVDT, capacitive, linear encoder, ultrasonic, laser triangulation
Angular displacement	Rotary potentiometer, RVDT, resolver, synchro, incremental/absolute encoders (resolution = $360°/2^n$ for n-bit absolute; incremental uses A/B quadrature, ×4 decoding)
Velocity	Tachogenerator (DC: V = kω), encoder frequency, moving-coil/magnet, Doppler
Acceleration	Seismic mass accelerometers (piezoelectric, piezoresistive, capacitive MEMS). Seismic instrument: accelerometer uses $\omega \ll \omega_n$; vibrometer (displacement) uses $\omega \gg \omega_n$
Force	Load cells (strain gauge), piezoelectric, LVDT with spring, proving ring
Torque	Strain gauges at 45° on a shaft, torsion bar with angle sensing, slip rings or telemetry
Pressure	Bourdon tube, diaphragm, bellows, manometer, piezoresistive/capacitive/piezoelectric sensors

4. Actuators

Hydraulic and pneumatic

- Pascal's law: pressure transmitted equally; F = pA. Cylinder extend force = pA_{bore}; retract force = $p(A_{bore} - A_{rod})$. Velocity = Q/A. Power = pQ.
- **Hydraulic:** incompressible oil, high force/power density, stiff, precise, but leaks and needs a pump and reservoir. Typical pressure 70–350 bar.
- **Pneumatic:** compressed air (~6–8 bar), fast, clean, cheap, compliant; poor position control due to compressibility.
- Components: pumps (gear, vane, piston), cylinders (single/double acting), motors, directional control valves (e.g. 4/3, 5/2: ports/positions), pressure relief valve, flow control valve, check valve, servo and proportional valves, accumulator.
- Hydraulic motor torque $T = pD/(2\pi)$ (D = displacement per rev); speed N = Q/D.

DC motors

- Back-EMF $E_b = k\varphi\omega$; $V = E_b + I_aR_a$; torque $T = k\varphi I_a$. Power: $E_bI_a = T\omega$.
- Permanent magnet / separately excited: $\omega = V/k\varphi - R_aT/(k\varphi)^2$ — linear drooping torque-speed line. Stall torque at ω = 0; no-load speed at T = 0. Max power at half the no-load speed.
- Series motor: $T \propto I^2$ (high starting torque, dangerous at no load). Shunt: nearly constant speed.
- Speed control: armature voltage (below base speed), field weakening (above base speed). PWM with an H-bridge for direction.
- Transfer function $\theta(s)/V(s) = K/[s(Js + b)(Ls + R) + K^2] \approx K_m/[s(\tau_m s + 1)]$.

Stepper motors

- Step angle = $360°/(2m \cdot N_r)$ for PM/hybrid (m = phases, N_r = rotor teeth or pole pairs; e.g. 2-phase hybrid with 50 teeth → 1.8°); variable reluctance: $(N_s - N_r)/(N_sN_r) \times 360°$.
- Steps/rev = 360°/step angle. Speed (rpm) = (step rate × step angle × 60)/360.
- Types: permanent magnet, variable reluctance, hybrid (typically 1.8°). Drive modes: full step, half step (doubles resolution), microstepping.

- Open-loop positioning; holding torque, detent torque, pull-in and pull-out torque curves; can lose steps or resonate.

Servo motors

- Motor + feedback sensor (encoder/resolver/potentiometer) + controller in a closed loop. DC servo, AC servo (BLDC/PMSM), RC servo (PWM: 1–2 ms pulse every 20 ms sets angle).
- BLDC: electronic commutation using Hall sensors; high efficiency, no brush wear.
- AC induction motor: synchronous speed $N_s = 120f/P$; slip $s = (N_s - N)/N_s$.

5. Engineering Mechanics

Free-body diagrams and equilibrium

- FBD: isolate the body, show all external forces and moments (weights, reactions, friction, applied loads).
- 2D equilibrium: $\Sigma F_x = 0$, $\Sigma F_y = 0$, $\Sigma M = 0$ (3 equations). 3D: 6 equations.
- Supports: roller (1 reaction), pin/hinge (2), fixed (3 incl. moment).
- **Lami's theorem** (3 concurrent forces): $F_1/\sin\alpha = F_2/\sin\beta = F_3/\sin\gamma$ (angle opposite each force). Two-force member: forces equal, opposite, collinear.
- Varignon: moment of resultant = sum of moments of components. Couple: moment independent of point.
- Virtual work: $\Sigma F \cdot \delta r = 0$ for equilibrium.

Friction

- Coulomb: $F \le \mu_s N$; sliding $F = \mu_k N$ ($\mu_k < \mu_s$). Angle of friction $\tan\varphi = \mu$ = angle of repose.
- Block on incline: slides if $\tan\theta > \mu$. Force up the incline $P = W(\sin\theta + \mu\cos\theta)$; to prevent sliding down $P = W(\sin\theta - \mu\cos\theta)$.
- Wedges, screw jack: effort $P = W\tan(\alpha + \varphi)$; self-locking if $\varphi > \alpha$; efficiency $= \tan\alpha/\tan(\alpha + \varphi)$.
- **Rolling friction:** resistance $F = W \cdot a/r$ (a = coefficient of rolling resistance, length units). Much smaller than sliding friction.
- **Belt friction:** $T_1/T_2 = e^{\mu\theta}$ (θ in radians, T_1 tight side). V-belt: $e^{\mu\theta/\sin(\beta/2)}$. Power $= (T_1 - T_2)v$. With centrifugal tension $T_c = mv^2$, max power when $T_c = T_{max}/3$. Open belt vs crossed belt lengths; crossed belt reverses direction.

Trusses and frames

- Truss: two-force members joined by pins, loads at joints. Simple (perfect) plane truss: $m = 2j - 3$. $m < 2j - 3 \rightarrow$ mechanism; $m > 2j - 3 \rightarrow$ statically indeterminate.
- **Method of joints:** equilibrium at each joint (2 equations, ≤2 unknowns). **Method of sections:** cut through ≤3 members, use 3 equations.
- Zero-force members: at an unloaded joint with two non-collinear members, both are zero; with three members where two are collinear, the third is zero.
- Frames/machines contain multi-force members; disassemble and apply equilibrium to each part.

Kinematics of rigid bodies (plane motion)

- Particle: $v = u + at$; $s = ut + \frac{1}{2}at^2$; $v^2 = u^2 + 2as$. Projectile: range $= u^2\sin 2\theta/g$; max height $= u^2\sin^2\theta/2g$; time $= 2u\sin\theta/g$.
- Circular: $v = \omega r$; $a_t = \alpha r$; $a_n = \omega^2 r = v^2/r$.
- Rigid body: $v_B = v_A + \omega \times r_{B/A}$; $a_B = a_A + \alpha \times r_{B/A} - \omega^2 r_{B/A}$.

- **Instantaneous centre (IC):** point of zero velocity; v = ω × (distance from IC). Rolling without slipping: $v_{centre} = \omega R$, contact point at rest, top point 2v.

Dynamics of rigid bodies

- $\Sigma F = m\,a_G$; $\Sigma M_G = I_G\alpha$. D'Alembert: add inertia force $-ma$ and inertia torque $-I\alpha$, then treat as statics.
- Moments of inertia: rod about centre $mL^2/12$, about end $mL^2/3$; disc $mR^2/2$; ring mR^2; solid sphere $2mR^2/5$. Parallel axis $I = I_G + md^2$.
- Work–energy: $KE = \tfrac{1}{2}mv_G^2 + \tfrac{1}{2}I_G\omega^2$. Rolling down incline: $a = g\sin\theta/(1 + k^2/R^2)$.
- Impulse–momentum: $\int F\,dt = \Delta(mv)$; angular momentum $H = I\omega$ conserved if no external torque. Coefficient of restitution $e = (v_2 - v_1)/(u_1 - u_2)$.

6. Programming (Python)

- **Flowcharts:** oval = start/end, rectangle = process, parallelogram = input/output, diamond = decision, arrows = flow. **Pseudocode:** structured plain-language steps (IF/ELSE, WHILE, FOR).
- **Data types:** int (arbitrary precision), float, complex, bool, str (immutable), list (mutable), tuple (immutable), set (unique, unordered), dict (key → value; keys must be hashable), NoneType.
- **Operators:** arithmetic + − * / (true division, always float) // (floor division) % (modulo; sign follows divisor: −7 % 3 = 2, −7 // 3 = −3) ** (power, right-associative: 2**3**2 = 512). Comparison, logical (and/or/not, short-circuit), bitwise (& | ^ ~ << >>; ~x = −x − 1), membership (in), identity (is).
- Precedence: ** > unary − > * / // % > + − > shifts > & > ^ > | > comparisons > not > and > or.
- **Conditionals:** if / elif / else; ternary x if cond else y; match–case (Python 3.10+) is the "case" construct.
- **Loops:** for x in iterable; while cond. range(start, stop, step) excludes stop. break exits the loop; continue skips to next iteration; a loop's else runs only if no break happened.
- **Functions:** def f(a, b=2, *args, **kwargs). Default arguments are evaluated once (mutable default pitfall). Variables assigned inside a function are local unless declared global/nonlocal. Functions are first-class; lambda for small anonymous functions.
- **Arrays/lists:** zero-indexed; negative index from end; slicing a[start:stop:step] (a[::-1] reverses). Lists are references: b = a aliases, b = a[:] or a.copy() makes a shallow copy. append O(1), insert/pop(0) O(n).
- **Dictionaries:** d[k], d.get(k, default), d.keys()/values()/items(); insertion-ordered (3.7+); average O(1) lookup.
- **Strings:** immutable; s[i], slicing, len, s.upper(), s.split(), ''.join(list), s.find(), s.replace(), f-strings. "ab" * 3 = "ababab".
- **Recursion:** base case + recursive case. Factorial, Fibonacci (naive is $O(2^n)$), Tower of Hanoi ($2^n - 1$ moves). Default recursion limit ~1000. Trace by drawing the call tree.

7. Data Structures

Structure	Key idea	Operations / complexity
Stack	LIFO	push/pop/peek O(1). Uses: recursion, expression evaluation, infix→postfix, balanced parentheses, undo
Queue	FIFO	enqueue/dequeue O(1). Circular queue: full when (rear+1) % n == front. Deque, priority queue. Uses: BFS, scheduling
Linked list	Nodes with pointers	Insert/delete at head O(1); search O(n); no random access. Singly, doubly, circular. Reverse in O(n); detect cycle with slow/fast pointers
Binary search tree	left < node < right	Search/insert/delete O(h): O(log n) balanced, O(n) skewed. Inorder traversal gives sorted order. Delete with 2 children: replace by inorder successor/predecessor
Binary heap	Complete tree; parent ≥ children (max-heap)	Array: children of i at 2i+1, 2i+2; parent $\lfloor (i-1)/2 \rfloor$. Insert/delete-max O(log n); peek O(1); build-heap O(n); heapsort O(n log n)
Graph	G = (V, E)	Adjacency matrix $O(V^2)$ space; adjacency list O(V + E). BFS/DFS O(V + E). Dijkstra O((V+E) log V) with heap; topological sort for DAGs; MST via Prim/Kruskal

- Tree facts: binary tree of height h has at most 2^(h+1) − 1 nodes; n nodes → n − 1 edges; leaves = internal nodes with 2 children + 1.
- Traversals: preorder (NLR), inorder (LNR), postorder (LRN), level order (BFS). Inorder + preorder (or postorder) uniquely determines a binary tree.
- Number of distinct BSTs with n keys = Catalan number C(2n, n)/(n + 1).
- Undirected graph: Σdegree = 2|E|; complete graph has n(n−1)/2 edges; tree on n vertices has n − 1 edges.

Part A.3 — Principles of Robotics and Automation

1. Robot Classification

- **Serial manipulator:** open kinematic chain, one path from base to end-effector. Large workspace, dexterous, but errors accumulate along the chain and stiffness/payload is lower. Forward kinematics easy, inverse kinematics hard (multiple solutions).
- **Parallel manipulator:** closed chains; the end-effector (platform) is connected to the base by several legs (e.g. Stewart–Gough platform 6-DOF, Delta robot 3-DOF translation). High stiffness, accuracy, speed, payload-to-weight ratio; small workspace. Inverse kinematics easy, forward kinematics hard.
- Other classifications: by drive (electric, hydraulic, pneumatic), by control (point-to-point, continuous path, servo/non-servo), by application (industrial, mobile, service), by generation/intelligence.

Geometrical configurations (first three joints → arm workspace)

Configuration	Joints	Workspace	Example use
Cartesian (gantry)	PPP	Rectangular box	Pick-and-place, CNC-like, 3D printers
Cylindrical	RPP	Hollow cylinder	Assembly, machine loading
Spherical (polar)	RRP	Partial hollow sphere	Welding, Stanford arm
Articulated (revolute/anthropomorphic)	RRR	Near-spherical, irregular	Painting, welding, most industrial arms
SCARA	RRP (two parallel vertical R axes)	Cylindrical-like	Fast planar assembly, compliant in XY, stiff in Z

2. Links, Joints, Coordinate Systems and DOF

- **Link:** rigid body. **Joint:** connection allowing relative motion.
- Lower pairs: revolute R (1 DOF), prismatic P (1), helical/screw H (1), cylindrical C (2), spherical S (3), planar E (3). Universal joint U (2).
- **DOF of a rigid body:** 3 in a plane, 6 in space (3 position + 3 orientation).
- **Grübler–Kutzbach (mobility):** planar $M = 3(n - 1) - 2j_1 - j_2$; spatial $M = 6(n - 1) - \Sigma(6 - f_i)$ (n = links incl. ground, j_1 = 1-DOF joints, j_2 = 2-DOF joints, f_i = DOF of joint i).
- A serial arm's DOF = sum of joint DOFs. 6 DOF are needed for arbitrary position and orientation; >6 is **redundant** (e.g. 7-DOF arms; avoid obstacles/singularities).
- Typical arm: 3 joints for position (arm) + 3 for orientation (wrist: roll–pitch–yaw / spherical wrist with intersecting axes).
- **Coordinate frames:** world/base frame, joint frames, tool (TCP) frame, user/workpiece frame. Joint space ($\theta_1 \ldots \theta_n$) vs task/Cartesian space (x, y, z, orientation).
- **Workspace:** reachable workspace (points reachable in at least one orientation) vs dexterous workspace (all orientations).

- **Singularity:** configuration where the Jacobian loses rank (det J = 0): loss of a DOF, infinite joint velocities needed; e.g. a fully stretched arm or two aligned wrist axes.

3. Rotation Matrices

2D

- R(θ) = [[cos θ, −sin θ], [sin θ, cos θ]]. Rotating a vector counter-clockwise by θ: p′ = R p.

3D elementary rotations

- $R_x(\theta)$ = [[1, 0, 0], [0, cθ, −sθ], [0, sθ, cθ]]
- $R_y(\theta)$ = [[cθ, 0, sθ], [0, 1, 0], [−sθ, 0, cθ]] (note the sign pattern)
- $R_z(\theta)$ = [[cθ, −sθ, 0], [sθ, cθ, 0], [0, 0, 1]]

Properties (SO(3))

- Columns of ${}^{A}R_{B}$ are the unit axes of frame B expressed in frame A.
- Orthogonal: $R^T R = I$, $R^{-1} = R^T$, det R = +1. Each row/column is a unit vector, mutually perpendicular. Preserves lengths and angles.
- Eigenvalues: 1, $e^{\pm i\theta}$. The eigenvector for λ = 1 is the rotation axis. trace R = 1 + 2 cos θ.
- **Composition:** ${}^{A}R_{C} = {}^{A}R_{B}\ {}^{B}R_{C}$. Rotations about the **current (moving)** frame → post-multiply; about the **fixed** frame → pre-multiply. Matrix multiplication is not commutative.
- **Euler angles** (ZYZ): $R = R_z(\varphi)R_y(\theta)R_z(\psi)$. **Roll–pitch–yaw** (fixed XYZ): $R = R_z(\alpha)R_y(\beta)R_x(\gamma)$. Both have singularities (gimbal lock).
- **Axis–angle (Rodrigues):** $R = I + \sin\theta\,[k]_x + (1-\cos\theta)[k]_x^2$. Quaternions avoid singularities: q = (cos θ/2, k sin θ/2).

4. Homogeneous Transformations

- T = [[R, d], [0 0 0, 1]] (4×4), combining rotation R and translation d. Point in homogeneous form $\tilde{p} = [x, y, z, 1]^T$; ${}^{A}\tilde{p} = {}^{A}T_{B}\ {}^{B}\tilde{p}$.
- Inverse: $T^{-1} = [[R^T, -R^T d], [0, 1]]$ (not simply the transpose).
- Chain: ${}^{0}T_{n} = {}^{0}T_{1}\ {}^{1}T_{2} \ldots {}^{n-1}T_{n}$. Same pre/post-multiplication rule as for rotations.
- Pure translation Trans(a, b, c) has R = I; pure rotation has d = 0.
- Free vectors (directions) use a 0 as the fourth component, so translation does not affect them.

5. Forward Kinematics

- Given joint variables, find end-effector position and orientation: ${}^{0}T_{n}(q)$.
- **Denavit–Hartenberg (standard) convention:** z_i along joint i+1 axis; x_i along the common normal from z_{i-1} to z_i. Four parameters per link:
  - θ_i — joint angle about z_{i-1} (variable for revolute)
  - d_i — offset along z_{i-1} (variable for prismatic)
  - a_i — link length along x_i
  - α_i — link twist about x_i
- ${}^{i-1}A_i = \mathrm{Rot}_z(\theta_i)\,\mathrm{Trans}_z(d_i)\,\mathrm{Trans}_x(a_i)\,\mathrm{Rot}_x(\alpha_i)$ = [[cθ, −sθcα, sθsα, a cθ], [sθ, cθcα, −cθsα, a sθ], [0, sα, cα, d], [0, 0, 0, 1]].

- **2-link planar arm** (lengths L_1, L_2): $x = L_1\cos\theta_1 + L_2\cos(\theta_1 + \theta_2)$; $y = L_1\sin\theta_1 + L_2\sin(\theta_1 + \theta_2)$; orientation $\varphi = \theta_1 + \theta_2$. Workspace: annulus with radii $|L_1 - L_2|$ and $L_1 + L_2$.
- **Inverse kinematics** (2-link): $\cos\theta_2 = (x^2 + y^2 - L_1^2 - L_2^2)/(2L_1L_2)$ (two solutions: elbow up/down); $\theta_1 = \text{atan2}(y, x) - \text{atan2}(L_2\sin\theta_2, L_1 + L_2\cos\theta_2)$.
- **Jacobian:** $\dot{x} = J(q)\,\dot{q}$ relates joint velocities to end-effector velocity; $\tau = J^T F$ relates end-effector force to joint torques. For the 2-link arm: $J = [[-L_1s_1 - L_2s_{12}, -L_2s_{12}], [L_1c_1 + L_2c_{12}, L_2c_{12}]]$, $\det J = L_1L_2\sin\theta_2$ (singular at $\theta_2 = 0, \pi$).

6. Robot Applications and Performance

- **Point-to-point (PTP) control:** only the end points matter, path between them is not controlled (spot welding, pick-and-place, machine loading, drilling). Often joint-interpolated motion.
- **Continuous path (CP) control:** the whole trajectory is controlled (arc welding, spray painting, gluing, deburring). Requires Cartesian interpolation (linear/circular) and coordinated joints.
- Trajectory planning: cubic polynomial (4 constraints: positions and velocities at both ends), quintic (adds accelerations), linear segments with parabolic blends (LSPB/trapezoidal velocity).
- Teaching/programming: lead-through, teach pendant, offline programming (simulation).
- Applications: material handling, palletizing, machine tending, welding, painting, assembly, inspection, medical, space, underwater.

End-effectors

- **Grippers:** mechanical (2/3-finger, parallel or angular jaws; friction grip $\mu \cdot n_f \cdot F_g \geq W \times g$-factor, n_f = contact surfaces, g-factor accounts for acceleration), vacuum (suction cups; $F = \Delta P \cdot A$; flat, smooth parts), magnetic (electromagnet or permanent; ferrous parts), adhesive, soft/compliant, multi-fingered hands.
- **Tools:** welding torch, spot-welding gun, spray gun, drill/spindle, screwdriver, laser/water-jet cutter.
- Tool changers; Remote Centre Compliance (RCC) device for peg-in-hole assembly.

Accuracy, repeatability, resolution

- **Resolution (control resolution):** smallest increment the controller can command; = range/2^n for an n-bit encoder/control word.
- **Accuracy:** how close the robot gets to a commanded target. Accuracy = ½ control resolution + ½ mechanical inaccuracy spread (3σ). Errors come from link deflection, gear backlash, calibration, thermal expansion.
- **Repeatability:** how close the robot returns to a previously taught point ($\pm 3\sigma$ of the spread). Industrial robots: repeatability (~±0.02–0.1 mm) is much better than absolute accuracy.
- Other specs: payload, reach, speed, DOF, work envelope, compliance.

7. Computer Integrated Manufacturing (CIM)

Automation in manufacturing

- **Fixed (hard) automation:** high volume, low variety, high initial cost (transfer lines). **Programmable:** batch production, reprogrammable (CNC, robots). **Flexible:** variety without lost changeover time (FMS).

- Production metrics: cycle time T_c; production rate $R_p = 1/T_p$; utilization; availability A = (MTBF − MTTR)/MTBF; manufacturing lead time MLT = $n_o(T_{su} + Q\,T_c + T_{no})$; WIP.
- Levels of automation: device → machine → cell/system → plant → enterprise. Ten strategies: specialization of operations, combined operations, simultaneous operations, integration of operations, increased flexibility, improved material handling and storage, on-line inspection, process control and optimization, plant operations control, CIM.
- Transfer line efficiency $E = T_c/(T_c + F{\cdot}T_d)$ (F = breakdown frequency, T_d = downtime). Line balancing: balance efficiency = $T_{wc}/(n \times T_s)$.

Programmable Logic Controllers (PLCs)

- Industrial computer: CPU, memory, input modules (sensors/switches), output modules (relay, transistor, triac), power supply, programming device; communication (Modbus, Profibus, EtherNet/IP).
- **Scan cycle:** read inputs → execute program → update outputs → housekeeping/communication. Scan time typically ms; an input shorter than one scan can be missed.
- IEC 61131-3 languages: Ladder Diagram (LD), Function Block Diagram (FBD), Structured Text (ST), Instruction List (IL), Sequential Function Chart (SFC).
- Ladder logic: series contacts = AND, parallel contacts = OR, NC contact = NOT. Seal-in (latch) circuit for start/stop: (Start + Motor)·Stop′ → Motor.
- Timers (TON on-delay, TOF off-delay, RTO retentive), counters (CTU, CTD), comparison/math, sequencers. Advantages over relays: reprogrammable, compact, diagnostics.

Automated material handling

- Industrial trucks, **AGVs** (automated guided vehicles: wire/tape/laser/vision guidance; towing, unit-load, forklift types) and AMRs, conveyors (roller, belt, chain, overhead trolley, cart-on-track), cranes and hoists, monorails, industrial robots.
- Principles: unit load, minimum handling, straight-line flow, standardization, gravity use. AGV system sizing: number of vehicles = total workload / available time per vehicle.

Automated storage and retrieval systems (AS/RS)

- Storage racks + storage/retrieval (S/R) machines in aisles + pickup/deposit (P&D) stations, under computer control. Types: unit load, mini-load, man-on-board, deep-lane, carousel storage (horizontal/vertical).
- Cycle time: single command $T_{sc} = 2\max(h/v_y, L/v_x) + 2T_{pd}$; dual command $T_{dc} = 2\max(1.5h/v_y, 1.5L/v_x) + 4T_{pd}$ (MH-10 approximation).
- Benefits: higher density, less labour, better inventory control, security.

Automated identification and data capture (AIDC)

- **Bar codes:** 1D linear (Code 39, Code 128, UPC/EAN) and 2D (stacked PDF417; matrix QR, Data Matrix). Read by laser/CCD scanners.
- **RFID:** tag (chip + antenna) and reader; passive (powered by reader) or active (battery); no line-of-sight needed, read/write, multiple tags at once; frequencies LF/HF/UHF.
- Others: magnetic stripes, OCR, machine vision, smart cards. AIDC reduces errors versus manual entry (~1 in 300 keystrokes vs ~1 in millions for bar codes).

Computer numerical control (CNC)

- NC: program of numerical instructions controls a machine tool. CNC: a dedicated computer stores and executes the part program. DNC: several machines served from a central computer.

- Components: part program, machine control unit (MCU), processing equipment. Axes: Z along spindle, X longest horizontal, Y by right-hand rule; rotary A, B, C about X, Y, Z.
- Motion control: point-to-point vs contouring (continuous path); interpolation linear (G01) and circular (G02 CW, G03 CCW). G00 rapid positioning; G90 absolute vs G91 incremental; G17/18/19 plane selection; G41/G42 cutter radius compensation; M03 spindle CW; M05 spindle stop; M06 tool change; M08 coolant on; M30 end of program.
- Control: open loop (stepper) vs closed loop (servo + encoder feedback). **Control resolution** (open loop with stepper and leadscrew): BLU = $p/(n_s \cdot r_g)$ (p = screw pitch, n_s = steps/rev, r_g = gear ratio). Pulse rate $f_p = v \cdot n_s \cdot r_g / p$.
- Machining centres with automatic tool changer (ATC), automatic pallet changer. APT language, CAM post-processing.

Single and multi-axis positioning systems

- Single axis: motor + leadscrew/ball screw (low friction, high efficiency, minimal backlash with preload) or belt/rack-and-pinion/linear motor + guideway + feedback (rotary encoder, linear scale).
- Multi-axis: stacked XY tables, gantries, CNC axes. Positioning accuracy is affected by resolution, mechanical errors (backlash, pitch error, thermal), and Abbe error.
- Ball screw: linear speed $v = N \cdot p$; thrust $F = 2\pi\eta T/p$.

Concurrent design and manufacturing planning

- **Concurrent (simultaneous) engineering:** design, manufacturing, quality, purchasing and service work in parallel teams instead of sequentially ("over the wall"). Shorter lead time, fewer changes, lower cost.
- **DFM / DFA** (design for manufacture / assembly): minimize part count, use standard parts, design for top-down assembly, self-locating features, avoid flexible parts; Boothroyd–Dewhurst DFA index.
- Design for automation: parts easy to feed and orient (symmetry or obvious asymmetry), avoid tangling, fewer fasteners.
- **Process planning:** manual or computer-aided (CAPP): variant (retrieve similar plan via group technology/coding, e.g. Opitz) or generative (logic creates a new plan).
- MRP (material requirements planning), MRP II, ERP; capacity planning; master production schedule; shop floor control; JIT/lean; group technology and cellular manufacturing.

Part B1 — Electrical Engineering

B1.1 Analog Circuits and Embedded Systems

Analog filters

- **Low-pass:** passes $f < f_c$. First-order RC: $H(s) = 1/(1 + sRC)$, $f_c = 1/(2\pi RC)$; −20 dB/decade roll-off per order; −3 dB and −45° at f_c.
- **High-pass:** $H(s) = sRC/(1 + sRC)$. **Band-pass:** passes $f_1 < f < f_2$; centre $f_0 = \sqrt{(f_1 f_2)}$, $Q = f_0/BW$. **Band-reject (notch):** rejects a band (e.g. 50 Hz hum; twin-T).
- Second-order standard form: LP $H = \omega_0^2/(s^2 + (\omega_0/Q)s + \omega_0^2)$; −40 dB/decade.
- Passive (RLC, no gain, loading problems) vs active (op-amp; gain, buffering, no inductors). Sallen–Key and multiple-feedback topologies.
- Approximations: Butterworth (maximally flat passband), Chebyshev (ripple, sharper roll-off), Bessel (linear phase), Elliptic (sharpest, ripple in both bands). Order n → roll-off 20n dB/decade.

Amplifiers: biasing, equivalent circuit, frequency response

- **BJT:** $I_C = \beta I_B$, $I_E = I_C + I_B$, $\alpha = \beta/(\beta + 1)$. Active region: BE forward, BC reverse. $V_{BE} \approx 0.7$ V (Si). Thermal voltage $V_T \approx 25$ mV at room temperature.
- **Biasing:** fixed bias (unstable), collector-feedback, **voltage-divider bias** (most stable; Thevenin: $V_{BB} = V_{CC}R_2/(R_1+R_2)$, $I_E \approx (V_{BB} - V_{BE})/R_E$ when $\beta R_E \gg R_B$). Stability factor $S = \partial I_C/\partial I_{CO}$; Q-point on the DC load line, middle for maximum symmetric swing.
- **Small-signal (hybrid-π):** $g_m = I_C/V_T$; $r_\pi = \beta/g_m$; $r_o = V_A/I_C$. h-parameters: $h_{ie} \approx r_\pi$, $h_{fe} \approx \beta$.
- CE: $A_v \approx -g_m(R_C \| R_L)$, inverting, medium R_{in}. With unbypassed R_E: $A_v \approx -R_C/R_E$. CB: non-inverting, low R_{in}, current gain ≈ 1. CC (emitter follower): $A_v \approx 1$, high R_{in}, low R_{out} (buffer).
- **MOSFET:** saturation $I_D = \tfrac{1}{2}k(V_{GS} - V_t)^2$; $g_m = 2I_D/(V_{GS} - V_t) = \sqrt{(2kI_D)}$. Saturation if $V_{DS} \geq V_{GS} - V_t$. CS ≈ CE, CG ≈ CB, CD (source follower) ≈ CC.
- **Frequency response:** low-frequency cut-off from coupling/bypass capacitors; high-frequency from junction/parasitic capacitances. **Miller effect:** $C_{in} = C_\mu(1 - A_v)$ — large for high-gain inverting stages. Gain–bandwidth product is roughly constant.
- Multistage: overall gain = product; overall bandwidth shrinks. Power amplifier classes: A (η ≤ 25%/50%), B (78.5%, crossover distortion), AB, C (tuned).

Feedback amplifiers

- Closed-loop gain $A_f = A/(1 + A\beta)$ (negative feedback). Desensitivity $D = 1 + A\beta$: gain sensitivity reduced by D, bandwidth increased by D, distortion and noise reduced.
- Topologies (sampling–mixing): voltage-series (series–shunt): $R_{in} \times D$, $R_{out} \div D$ — ideal voltage amplifier. Current-series: $R_{in} \times D$, $R_{out} \times D$ (transconductance). Voltage-shunt: $R_{in} \div D$, $R_{out} \div D$ (transresistance). Current-shunt: $R_{in} \div D$, $R_{out} \times D$ (current amplifier). Rule: series mixing raises R_{in}, shunt lowers; voltage sampling lowers R_{out}, current sampling raises.

Operational amplifiers

- **Ideal:** infinite open-loop gain, $R_{in} = \infty$, $R_{out} = 0$, infinite bandwidth, CMRR = ∞. With negative feedback: virtual short ($V_+ = V_-$) and no input current.

- **Non-ideal parameters:** input offset voltage/current, bias current, CMRR = A_d/A_{cm}, slew rate (max dV_o/dt; full-power bandwidth $f_{max} = SR/(2\pi V_m)$), gain–bandwidth product (unity-gain frequency f_T: closed-loop BW = f_T/A_{CL}), output saturation.

Circuit	Output
Inverting	$V_o = -(R_f/R_1)V_i$; $R_{in} = R_1$
Non-inverting	$V_o = (1 + R_f/R_1)V_i$; $R_{in} \approx \infty$
Voltage follower	$V_o = V_i$ (buffer)
Summing (inverting)	$V_o = -R_f(V_1/R_1 + V_2/R_2 + \ldots)$
Difference	$V_o = (R_2/R_1)(V_2 - V_1)$ with matched resistors
Integrator	$V_o = -(1/RC)\int V_i dt$ (add R across C to limit DC gain)
Differentiator	$V_o = -RC\, dV_i/dt$ (noisy at high frequency)
Instrumentation amp (3 op-amp)	$V_o = (1 + 2R/R_G)(R_3/R_2)(V_2 - V_1)$; high CMRR and input impedance
Log / antilog	Diode/BJT in feedback; $V_o \propto -V_T \ln(V_i/RI_s)$
Precision rectifier	Diode inside feedback loop removes the 0.7 V drop
Comparator	Open loop; output $\pm V_{sat}$ depending on sign of $V_+ - V_-$

Oscillators

- **Barkhausen criterion:** loop gain $|A\beta| = 1$ and total phase shift = 0° (or 360°). In practice $|A\beta|$ slightly > 1 at start-up.
- **RC phase-shift:** three RC sections give 180°; $f = 1/(2\pi RC\sqrt{6})$; amplifier gain ≥ 29.
- **Wien bridge:** $f = 1/(2\pi RC)$; $\beta = 1/3$ at f_0, so non-inverting gain ≥ 3 ($R_f = 2R_1$).
- LC: **Colpitts** $f = 1/(2\pi\sqrt{L\, C_{eq}})$, $C_{eq} = C_1C_2/(C_1 + C_2)$. **Hartley** $f = 1/(2\pi\sqrt{L_{eq}C})$, $L_{eq} = L_1 + L_2 + 2M$. **Crystal:** very stable frequency (high Q), series and parallel resonance.
- Relaxation oscillators (square/triangular): op-amp astable, $T = 2RC \ln[(1 + \beta)/(1 - \beta)]$, $\beta = R_2/(R_1 + R_2)$.

VCOs and timers

- **VCO:** output frequency proportional to control voltage (e.g. 566). Used in PLLs (phase detector + loop filter + VCO), FM, frequency synthesis.
- **555 timer:** two comparators (thresholds 2/3 V_{CC} and 1/3 V_{CC}), SR flip-flop, discharge transistor.
- Monostable: pulse width $T = 1.1RC$.
- Astable: $t_H = 0.693(R_A + R_B)C$; $t_L = 0.693R_BC$; $f = 1.44/[(R_A + 2R_B)C]$; duty cycle = $(R_A + R_B)/(R_A + 2R_B) > 50\%$ (add a diode across R_B for ≤ 50%).

Schmitt trigger

- Comparator with positive feedback → two thresholds (hysteresis), giving noise immunity and squaring slow signals.
- Inverting Schmitt (feedback divider R_1 to ground, R_2 to output): $V_{UT} = +V_{sat}R_1/(R_1 + R_2)$, $V_{LT} = -V_{sat}R_1/(R_1 + R_2)$; hysteresis width $V_H = V_{UT} - V_{LT}$.

Sample-and-hold

- Switch (MOSFET) + hold capacitor + buffer op-amps. Samples input during sample mode and holds constant during ADC conversion.

- Specs: acquisition time, aperture time/jitter, droop rate ($dV/dt = I_{leak}/C$), feedthrough. Larger C: less droop but slower acquisition.

A/D and D/A converters

- **Resolution:** LSB = $V_{FS}/2^n$ (some texts use $V_{FS}/(2^n - 1)$). Quantization error ±½ LSB; SQNR ≈ 6.02n + 1.76 dB.
- **DAC:** weighted-resistor (needs wide resistor range: R, 2R, 4R … $2^{n-1}R$) and **R-2R ladder** (only two resistor values; $V_o = -V_{ref}(R_f/R)\cdot\Sigma b_i 2^i/2^n$). Specs: settling time, monotonicity, linearity (INL/DNL).

ADC type	Conversion time	Notes
Flash (parallel)	1 clock	Fastest; needs $2^n - 1$ comparators
Successive approximation (SAR)	n clocks	Binary search; common in microcontrollers
Counter (digital ramp)	Up to 2^n clocks	Slow, variable time
Tracking	Fast once locked	Up/down counter follows the input
Dual-slope (integrating)	Up to 2^{n+1} clocks	Slow, very accurate, rejects noise (DMMs); result independent of RC
Sigma-delta	Oversampling	High resolution, audio/instrumentation

Switched-mode power supplies (SMPS)

- Transistor switched at high frequency with duty cycle D, energy stored in L/C. Efficiency 80–95% (vs linear regulators, $\eta \approx V_o/V_i$). Smaller magnetics at higher frequency; more EMI and ripple.
- Continuous conduction mode (CCM) steady-state ratios: **Buck** $V_o = D\cdot V_i$; **Boost** $V_o = V_i/(1 - D)$; **Buck-boost** $V_o = -D\cdot V_i/(1 - D)$ (inverted); **Flyback** (isolated buck-boost) $V_o = V_i(N_2/N_1)\cdot D/(1 - D)$; forward converter (isolated buck).
- Derivation: volt-second balance on the inductor (average inductor voltage = 0) and charge balance on the capacitor.
- Buck ripple: $\Delta I_L = (V_i - V_o)D/(fL)$; $\Delta V_o = \Delta I_L/(8fC)$.

Microprocessors and microcontrollers

- **Microprocessor:** CPU only on chip (ALU, registers, control unit); memory and I/O external — general-purpose computing. **Microcontroller:** CPU + RAM + ROM/Flash + I/O ports + timers + ADC + serial interfaces on one chip — embedded control.
- Architectures: Von Neumann (shared bus for code and data) vs Harvard (separate; most MCUs). CISC (8085/8051, x86) vs RISC (ARM, AVR, PIC).
- **CPU:** ALU, accumulator, general registers, program counter (address of next instruction), stack pointer, instruction register, flags (carry, zero, sign, parity, auxiliary carry, overflow). Fetch–decode–execute cycle; machine cycles and T-states.
- **Memory:** ROM/EPROM/EEPROM/Flash (non-volatile program), SRAM (fast, no refresh) and DRAM (refresh). Address lines n → 2^n locations (e.g. 16 lines → 64 KB). Memory map; address decoding with decoders/logic.
- **I/O ports:** parallel GPIO with direction registers; 8051 has four 8-bit ports (P0–P3). Serial: UART (asynchronous, start/stop bits, baud rate), SPI (full duplex, master/slave, MOSI/MISO/SCK/SS), I²C (two-wire SDA/SCL, addressed, multi-master), CAN (vehicles/industrial).

- **Timers/counters:** count internal clock pulses (timer) or external events (counter). 8051 timer modes 0–3 (13-bit, 16-bit, 8-bit auto-reload, split). Delay = count × machine-cycle time (8051: 12 clock periods per machine cycle). Used for PWM, input capture, baud-rate generation, watchdog.

Interrupt handling

- Interrupt: external/internal event suspends the main program; CPU finishes the current instruction, pushes PC (and status) onto the stack, jumps to the ISR via the vector table, then returns (RETI/IRET) and restores context.
- Polling vs interrupts (interrupts give faster response and free the CPU). Maskable vs non-maskable (NMI, e.g. 8085 TRAP). Vectored vs non-vectored. Edge vs level triggering. Priority and nesting.
- 8085 interrupts priority: TRAP > RST 7.5 > RST 6.5 > RST 5.5 > INTR. 8051: five sources (INT0, TF0, INT1, TF1, serial) with IE and IP registers.
- Good ISR practice: short, no blocking calls, mark shared variables volatile, disable interrupts around critical sections.

Memory and I/O interfacing

- Memory-mapped I/O (I/O shares memory address space; all memory instructions usable) vs I/O-mapped / isolated I/O (separate IN/OUT instructions, 8085: 256 ports).
- Address decoding: full vs partial (partial causes aliasing/foldback). Chip select from address decoder (e.g. 74LS138).
- Data transfer: programmed I/O, interrupt-driven, DMA (direct memory access; CPU releases buses via HOLD/HLDA for block transfers).
- Peripheral chips: 8255 PPI, 8253/8254 timer, 8259 interrupt controller, 8251 USART, 8237 DMA controller. Interfacing keypads, 7-segment/LCD displays, ADC/DAC, motors (through drivers, H-bridge, opto-isolation).

Data acquisition systems

- Chain: sensor → signal conditioning (amplify, filter, isolate) → anti-aliasing low-pass filter → analog multiplexer (multiple channels) → sample-and-hold → ADC → microcontroller/computer → storage/display.
- Sampling rate ≥ 2 × highest frequency (Nyquist); in practice 5–10×. Aliasing if undersampled. Throughput for N multiplexed channels: per-channel rate = total rate / N.
- Specs: number of channels, resolution, sampling rate, input range, accuracy, isolation. Single-ended vs differential inputs (differential rejects common-mode noise).

B1.2 Signals and Systems

Signals

- Continuous-time x(t) and discrete-time x[n]. Periodic: $x(t + T) = x(t)$. Sum of periodic signals is periodic if the ratio of periods is rational; period = LCM. $x[n] = \cos(\omega_0 n)$ is periodic only if $\omega_0/2\pi$ is rational.
- Even/odd decomposition: $x_e = \tfrac{1}{2}[x(t) + x(-t)]$, $x_o = \tfrac{1}{2}[x(t) - x(-t)]$.
- Energy $E = \int |x|^2 dt$ (finite ⇒ energy signal, P = 0); power $P = \lim (1/T)\int |x|^2 dt$ (finite non-zero ⇒ power signal, E = ∞).
- Standard signals: unit step u(t), impulse δ(t) ($\int \delta = 1$, sifting $\int x(t)\delta(t - t_0)dt = x(t_0)$, $\delta(at) = \delta(t)/|a|$), ramp r(t) = t u(t), rect, $\text{sinc}(x) = \sin(\pi x)/(\pi x)$, complex exponential e^{st}.

Shifting and scaling

- $x(t - t_0)$: delay (shift right) by t_0. $x(at)$: compress if $|a| > 1$, expand if $|a| < 1$. $x(-t)$: time reversal.
- For $y(t) = x(at - b)$: shift first by b, then scale by a (i.e. $x(a(t - b/a))$). Check with key points.
- Discrete: $x[2n]$ (decimation) loses samples; $x[n/2]$ (interpolation) defined only for even n.

System properties

- **Linear:** additivity + homogeneity (superposition). $y = ax + b$ with $b \neq 0$ is not linear (incrementally linear).
- **Time-invariant:** a delayed input gives the same delayed output. $y(t) = x(2t)$ or $y(t) = t\,x(t)$ are time-variant.
- **Causal:** output depends only on present and past inputs; LTI causal $\Leftrightarrow h(t) = 0$ for $t < 0$.
- **Memoryless (static):** output depends only on the present input; LTI $\Leftrightarrow h(t) = k\delta(t)$.
- **BIBO stable:** bounded input → bounded output; LTI $\Leftrightarrow \int|h(t)|dt < \infty$ ($\Sigma|h[n]| < \infty$).
- **Invertible:** distinct inputs → distinct outputs.
- LTI output: convolution $y(t) = x(t) * h(t) = \int x(\tau)h(t - \tau)d\tau$; $y[n] = \Sigma x[k]h[n - k]$. Commutative, associative, distributive. Cascade: $h_1 * h_2$; parallel: $h_1 + h_2$. Length of discrete linear convolution = $N + M - 1$. Eigenfunction: $e^{st} \rightarrow H(s)e^{st}$.

Fourier series

- CT periodic: $x(t) = \Sigma\, c_k\, e^{jk\omega_0 t}$, $c_k = (1/T)\int_T x(t)\, e^{-jk\omega_0 t}dt$. Real $x \Rightarrow c_{-k} = c_k^*$. Trigonometric form: $a_0 + \Sigma(a_k\cos + b_k\sin)$.
- Symmetry: even → cosine terms only; odd → sine terms only; half-wave symmetry $x(t \pm T/2) = -x(t)$ → odd harmonics only.
- Parseval: $P = \Sigma|c_k|^2$. Square wave has odd harmonics decaying as $1/k$; triangle wave as $1/k^2$. Gibbs phenomenon: ~9% overshoot at discontinuities.
- DT Fourier series: x[n] periodic with N; coefficients are also periodic with N (only N distinct).

Sampling theorem (Shannon)

- A band-limited signal with maximum frequency f_m can be perfectly recovered from samples if $f_s \geq 2f_m$ (Nyquist rate = $2f_m$; Nyquist interval = $1/2f_m$). Reconstruct with an ideal low-pass filter (sinc interpolation), cut-off between f_m and $f_s - f_m$.
- Sampled spectrum: copies of X(f) repeated every f_s. If $f_s < 2f_m$ → aliasing: a tone at f appears at $|f - k\, f_s|$ folded into $[0, f_s/2]$.
- For a product of signals, bandwidths add; for $x^2(t)$, bandwidth doubles. Band-pass sampling: $f_s = 2f_H/k$.

Fourier transform

- $X(\omega) = \int x(t)e^{-j\omega t}dt$; $x(t) = (1/2\pi)\int X(\omega)e^{j\omega t}d\omega$.
- Properties: linearity; time shift $x(t - t_0) \leftrightarrow e^{-j\omega t_0}X$; frequency shift $e^{j\omega_0 t}x \leftrightarrow X(\omega - \omega_0)$; scaling $x(at) \leftrightarrow (1/|a|)X(\omega/a)$; duality; differentiation $\leftrightarrow j\omega X$; integration $\leftrightarrow X/j\omega + \pi X(0)\delta(\omega)$; convolution $\leftrightarrow$ product; product $\leftrightarrow (1/2\pi)$ convolution; modulation $x\cos\omega_0 t \leftrightarrow \frac{1}{2}[X(\omega - \omega_0) + X(\omega + \omega_0)]$; Parseval $\int|x|^2dt = (1/2\pi)\int|X|^2d\omega$.
- Real and even $x \Rightarrow X$ real and even; real and odd $\Rightarrow X$ imaginary and odd. $X(0) = \int x\,dt$; $x(0) = (1/2\pi)\int X\,d\omega$.
- Pairs: $\delta \leftrightarrow 1$; $1 \leftrightarrow 2\pi\delta(\omega)$; $e^{-at}u(t) \leftrightarrow 1/(a + j\omega)$; $\text{rect}(t/T) \leftrightarrow T\,\text{sinc}(\omega T/2\pi)$; sinc $\leftrightarrow$ rect; $e^{-a|t|} \leftrightarrow 2a/(a^2 + \omega^2)$; Gaussian $\leftrightarrow$ Gaussian; $u(t) \leftrightarrow \pi\delta(\omega) + 1/j\omega$; sgn $\leftrightarrow 2/j\omega$.

- DTFT $X(e^{j\omega}) = \Sigma x[n]e^{-j\omega n}$ is periodic in ω with period 2π. DFT: N-point sampled DTFT; FFT computes it in O(N log N).

Laplace transform

- $X(s) = \int x(t)e^{-st}dt$ with region of convergence (ROC). Right-sided signals: ROC to the right of the rightmost pole; left-sided: left of the leftmost pole; two-sided: strip.
- LTI system H(s): causal ⇒ ROC is a right half-plane; stable ⇒ ROC includes the jω-axis; causal and stable ⇒ all poles in the LHP.
- Fourier transform = Laplace on s = jω (if ROC includes jω-axis). Properties as in the mathematics section (shift, derivative, convolution, initial/final value theorems).

Z-transform

- $X(z) = \Sigma x[n]z^{-n}$. Pairs: $\delta[n] \leftrightarrow 1$; $u[n] \leftrightarrow z/(z-1)$, $|z| > 1$; $a^n u[n] \leftrightarrow z/(z-a)$, $|z| > |a|$; $-a^n u[-n-1] \leftrightarrow z/(z-a)$, $|z| < |a|$; $n\,a^n u[n] \leftrightarrow az/(z-a)^2$.
- Properties: shift $x[n-k] \leftrightarrow z^{-k}X$; $a^n x \leftrightarrow X(z/a)$; $n\,x \leftrightarrow -z\,dX/dz$; convolution ↔ product; initial value $x[0] = \lim_{z\to\infty} X$; final value $x[\infty] = \lim_{z\to 1}(z-1)X$.
- ROC: no poles inside; causal ⇒ outside the outermost pole; stable ⇒ contains the unit circle; causal and stable ⇒ all poles inside the unit circle. DTFT = X(z) on $|z| = 1$. s-plane to z-plane: $z = e^{sT}$.
- Difference equation $y[n] - a\,y[n-1] = x[n] \rightarrow H(z) = 1/(1 - az^{-1})$. FIR (no feedback, always stable, can be linear phase) vs IIR (feedback poles).

RMS and average values

Waveform (peak V_m)	Average	RMS	Form factor (RMS/avg)
Sine (full cycle)	0	$V_m/\sqrt{2}$	—
Full-wave rectified sine	$2V_m/\pi$	$V_m/\sqrt{2}$	1.11
Half-wave rectified sine	V_m/π	$V_m/2$	1.57
Square ($\pm V_m$)	0	V_m	1 (half-cycle avg)
Triangle / sawtooth	0 (or $V_m/2$ if $0..V_m$)	$V_m/\sqrt{3}$	1.155
Pulse train, duty D, $0..V_m$	$D{\cdot}V_m$	$\sqrt{D}{\cdot}V_m$	$1/\sqrt{D}$

- General: $V_{avg} = (1/T)\int v\,dt$; $V_{rms} = \sqrt{[(1/T)\int v^2 dt]}$. For a sum of a DC and harmonics: $V_{rms} = \sqrt{(V_{dc}^2 + V_{1,rms}^2 + V_{2,rms}^2 + \ldots)}$. Crest factor = peak/RMS (√2 for a sine).

B1.3 Control Systems

Modelling and representation

- Mechanical translational: $F = M\ddot{x} + B\dot{x} + Kx$. Rotational: $T = J\ddot{\theta} + B\dot{\theta} + K\theta$. Electrical RLC.
- Analogies: force–voltage (M ↔ L, B ↔ R, K ↔ 1/C, x ↔ q) and force–current (M ↔ C, B ↔ 1/R, K ↔ 1/L).
- Gear trains: $\theta_1/\theta_2 = N_2/N_1$; $T_2/T_1 = N_2/N_1$; load inertia reflected to the motor side: $J_{eq} = J_m + J_L(N_1/N_2)^2$.
- State space: $\dot{x} = Ax + Bu$, $y = Cx + Du$; $G(s) = C(sI - A)^{-1}B + D$. Eigenvalues of A = poles. State transition matrix $\Phi(t) = e^{At} = L^{-1}\{(sI - A)^{-1}\}$. Controllability: rank[B AB … $A^{n-1}B$] = n; observability: rank[C; CA; …; CA^{n-1}] = n.

Feedback principle and transfer function

- Transfer function = L{output}/L{input} with zero initial conditions; = Laplace transform of the impulse response. Poles: roots of denominator; zeros: roots of numerator. Order = highest power of s in the denominator.
- Closed loop (negative feedback): T(s) = G/(1 + GH). Characteristic equation: 1 + G(s)H(s) = 0.
- Feedback reduces sensitivity to parameter changes ($S^T_G = 1/(1 + GH)$), rejects disturbances, can improve speed and stabilize, but can also destabilize and adds cost/complexity; it reduces overall gain.
- Open-loop: simple, no stability issue, but cannot correct disturbances.

Block diagrams and signal flow graphs

- Series: G_1G_2. Parallel: $G_1 \pm G_2$. Feedback loop: $G/(1 \pm GH)$. Moving a summing point ahead of a block: divide by G; behind: multiply by G. Moving a take-off point ahead: multiply by G; behind: divide by G.
- **Mason's gain formula:** $T = (1/\Delta)\Sigma P_k\Delta_k$. $\Delta = 1 -$ (sum of individual loop gains) + (sum of products of two non-touching loops) − (three non-touching) + … $\Delta_k = \Delta$ with the loops touching the k-th forward path removed.

Transient response

- First order $G = K/(\tau s + 1)$: step response $K(1 - e^{-t/\tau})$; 63.2% at $t = \tau$; settling time ≈ 4τ (2%); rise time (10–90%) = 2.2τ.
- Second order: $G = \omega_n^2/(s^2 + 2\zeta\omega_n s + \omega_n^2)$; poles $-\zeta\omega_n \pm j\omega_n\sqrt{(1 - \zeta^2)}$. $\zeta = 0$ undamped, $0 < \zeta < 1$ underdamped, 1 critical, > 1 overdamped.
- Underdamped step specs: damped frequency $\omega_d = \omega_n\sqrt{(1 - \zeta^2)}$; peak time $t_p = \pi/\omega_d$; percent overshoot $M_p = e^{-\pi\zeta/\sqrt{(1-\zeta^2)}} \times 100\%$; settling time $t_s = 4/(\zeta\omega_n)$ (2%) or $3/(\zeta\omega_n)$ (5%); rise time $t_r = (\pi - \theta)/\omega_d$, $\theta = \cos^{-1}\zeta$. Number of oscillations ≈ $t_s/(2\pi/\omega_d)$.
- Dominant poles: poles closest to the jω-axis dominate; poles > 5× further left can be neglected.

Steady-state error

- Unity feedback: $e_{ss} = \lim_{s\to 0} sR(s)/(1 + G(s))$. Type = number of poles of G(s)H(s) at the origin.
- Error constants: $K_p = \lim G(s)$; $K_v = \lim sG(s)$; $K_a = \lim s^2G(s)$ (as $s \to 0$).

Input	Type 0	Type 1	Type 2
Step A	$A/(1 + K_p)$	0	0
Ramp A t	∞	A/K_v	0
Parabola $At^2/2$	∞	∞	A/K_a

Routh–Hurwitz criterion

- Necessary: all coefficients of the characteristic polynomial present and of the same sign.
- Build the Routh array; number of sign changes in the first column = number of roots in the RHP. Stable iff no sign changes.
- Special cases: zero in first column (replace with $\varepsilon \to 0^+$); entire row zero (auxiliary polynomial A(s) from the row above, use dA/ds; indicates symmetric roots, e.g. on the jω-axis → marginal stability; roots of A(s) give the oscillation frequency).

- Second order: stable iff all coefficients > 0. Third order $a_0s^3 + a_1s^2 + a_2s + a_3$: stable iff all > 0 and $a_1a_2 > a_0a_3$. Use to find K range for stability and $K_{marginal}$.

Nyquist criterion

- Plot $G(j\omega)H(j\omega)$ for ω from $-\infty$ to ∞ (go around RHP poles/zeros on the jω-axis with small semicircles).
- **Z = P − N** (N = counter-clockwise encirclements of −1 + j0; P = open-loop RHP poles; Z = closed-loop RHP poles). Stable iff Z = 0, i.e. N = P. (With clockwise counting: Z = P + N.)
- Open-loop stable (P = 0) ⇒ stable iff no encirclement of −1.
- **Gain margin** GM = $1/|GH(j\omega_{pc})|$ at the phase-crossover frequency (phase = −180°); in dB: $-20\log|GH(j\omega_{pc})|$. **Phase margin** PM = 180° + $\angle GH(j\omega_{gc})$ at the gain-crossover frequency ($|GH| = 1$). Stable (minimum-phase) when both positive. PM ≈ 100ζ for $\zeta < 0.7$.

Bode plots

- Magnitude in dB ($20\log|G|$) and phase versus $\log\omega$.
- Gain K: flat $20\log K$, 0° (180° if negative). Pole at origin 1/s: −20 dB/dec through 0 dB at $\omega = 1$, −90°. Zero at origin: +20 dB/dec, +90°.
- Simple pole $1/(1 + s/a)$: 0 dB until $\omega = a$ then −20 dB/dec (actual −3 dB at the corner); phase 0 → −90° (−45° at a). Simple zero: +20 dB/dec, 0 → +90°.
- Quadratic pole: −40 dB/dec after ω_n, phase 0 → −180°; resonant peak $M_r = 1/(2\zeta\sqrt{(1 - \zeta^2)})$ at $\omega_r = \omega_n\sqrt{(1 - 2\zeta^2)}$ (for $\zeta < 0.707$).
- Initial slope reveals the system type (−20 × type dB/dec). Final slope = −20 × (poles − zeros). Non-minimum phase (RHP zero) shows extra phase lag for the same magnitude. Transport delay e^{-sT}: magnitude 1, phase $-\omega T$.
- Bandwidth of a second-order system: $\omega_b = \omega_n\sqrt{[(1 - 2\zeta^2) + \sqrt{(4\zeta^4 - 4\zeta^2 + 2)}]}$.

Root locus

- Locus of closed-loop poles as K varies 0 → ∞, from $1 + KG(s)H(s) = 0$. Angle condition: $\angle GH = \pm180°(2q + 1)$; magnitude condition: $|KGH| = 1$ (gives K at a point).
- Rules: starts at open-loop poles (K = 0), ends at zeros (K = ∞) or infinity. Number of branches = number of poles. Symmetric about the real axis. Real-axis segment is on the locus if an odd number of poles + zeros lie to its right.
- Asymptotes: n − m of them, angles $(2q + 1)180°/(n - m)$; centroid $\sigma = (\Sigma\text{poles} - \Sigma\text{zeros})/(n - m)$.
- Break-away/break-in points: $dK/ds = 0$. jω-axis crossings: Routh array. Angle of departure from a complex pole: 180° − Σ(angles from other poles) + Σ(angles from zeros); arrival at a complex zero analogously.
- Adding a pole pushes the locus right (less stable); adding a zero pulls it left (more stable).

Compensators

- **Lead** $G_c = (1 + aTs)/(1 + Ts)$, $a > 1$ (zero closer to origin than the pole). Adds phase lead: $\varphi_{max} = \sin^{-1}[(a - 1)/(a + 1)]$ at $\omega_m = 1/(T\sqrt{a})$. Improves PM, transient response and bandwidth (high-pass, amplifies noise). Approximates PD.
- **Lag** $G_c = (1 + aTs)/(1 + Ts)$, $a < 1$ (pole closer to origin). Increases low-frequency gain → reduces steady-state error; place well below crossover to avoid phase loss; reduces bandwidth (slower). Approximates PI.
- **Lead–lag:** combines both — lead for transient response, lag for steady-state accuracy.

P, PI, PID controllers

- $u(t) = K_p e + K_i \int e\,dt + K_d\, de/dt$; $G_c(s) = K_p + K_i/s + K_d s$.
- **P:** faster response, reduces but does not eliminate steady-state error; high gain → overshoot/instability.
- **I (PI):** adds a pole at the origin → raises system type by 1 → eliminates steady-state error to a step; can increase overshoot and reduce stability; integral wind-up (use anti-windup).
- **D (PD):** adds a zero → anticipates error, adds damping, reduces overshoot and settling time, improves stability; amplifies noise (use a filtered derivative); no effect on steady-state error.
- **PID:** combines all three. Ziegler–Nichols (ultimate-gain method): $K_p = 0.6K_u$, $T_i = P_u/2$, $T_d = P_u/8$; P only: $0.5K_u$; PI: $0.45K_u$, $T_i = P_u/1.2$.

Part B2 — Mechanical Engineering

B2.1 Mechanics of Materials

Stress and strain

- Normal stress $\sigma = P/A$; shear stress $\tau = V/A$. Normal strain $\varepsilon = \delta L/L$; shear strain γ (radians).
- Hooke's law: $\sigma = E\varepsilon$; $\tau = G\gamma$. Axial deformation $\delta = PL/(AE)$; for stepped bars sum each segment; bar under own weight $\delta = \rho g L^2/(2E) = WL/(2AE)$.
- Stress–strain curve (mild steel): proportional limit, elastic limit, upper/lower yield points, ultimate strength, fracture. Ductile (large plastic strain, cup-and-cone fracture) vs brittle (cast iron, little plastic strain).
- Factor of safety = failure stress/allowable stress. Strain energy $U = \sigma^2 V/(2E) = P^2 L/(2AE)$. Modulus of resilience $\sigma_y^2/(2E)$. Toughness = area under the curve.
- Suddenly applied load: stress = 2× gradually applied. Impact load from height h: $\sigma = (P/A)[1 + \sqrt{(1 + 2AEh/(PL))}]$.
- Composite bars in parallel: same strain, load shared in proportion to AE.

Elastic constants and Poisson's ratio

- Poisson's ratio $\nu = -\varepsilon_{lateral}/\varepsilon_{axial}$; $0 \le \nu \le 0.5$ (0.5 incompressible, rubber; ~0.3 steel; ~0 cork).
- $E = 2G(1 + \nu) = 3K(1 - 2\nu)$; $E = 9KG/(3K + G)$.
- Volumetric strain $\varepsilon_v = \varepsilon_x + \varepsilon_y + \varepsilon_z$; for uniaxial: $\varepsilon(1 - 2\nu)$. Bulk modulus $K = -p/\varepsilon_v$.
- Generalized Hooke: $\varepsilon_x = [\sigma_x - \nu(\sigma_y + \sigma_z)]/E$.

Mohr's circle — plane stress

- Stresses on a plane at θ: $\sigma_\theta = (\sigma_x + \sigma_y)/2 + [(\sigma_x - \sigma_y)/2]\cos 2\theta + \tau_{xy}\sin 2\theta$; $\tau_\theta = -[(\sigma_x - \sigma_y)/2]\sin 2\theta + \tau_{xy}\cos 2\theta$.
- Principal stresses $\sigma_{1,2} = (\sigma_x + \sigma_y)/2 \pm \sqrt{\{[(\sigma_x - \sigma_y)/2]^2 + \tau_{xy}^2\}}$; principal planes $\tan 2\theta_p = 2\tau_{xy}/(\sigma_x - \sigma_y)$; zero shear on principal planes.
- Max in-plane shear $\tau_{max} = (\sigma_1 - \sigma_2)/2$ = radius of Mohr's circle, on planes at 45° to principal planes (normal stress = average there). Absolute max shear (3D) = $(\sigma_{max} - \sigma_{min})/2$ including $\sigma_3 = 0$.
- Mohr's circle: centre $((\sigma_x + \sigma_y)/2, 0)$, radius R as above; an angle θ on the element = 2θ on the circle. $\sigma_x + \sigma_y$ is invariant.
- Pure shear: $\sigma_1 = \tau$, $\sigma_2 = -\tau$ at 45°. Uniaxial σ: $\tau_{max} = \sigma/2$ at 45°.

Plane strain

- Same transformation with $\sigma \to \varepsilon$ and $\tau \to \gamma/2$. Principal strains $\varepsilon_{1,2} = (\varepsilon_x + \varepsilon_y)/2 \pm \sqrt{\{[(\varepsilon_x - \varepsilon_y)/2]^2 + (\gamma_{xy}/2)^2\}}$. Max shear strain $\gamma_{max} = \varepsilon_1 - \varepsilon_2$.

Thin cylinders and spheres (t < d/20)

- Hoop (circumferential) $\sigma_h = pd/(2t)$; longitudinal $\sigma_l = pd/(4t) = \sigma_h/2$. Max in-plane shear = $pd/(8t)$; absolute max = $pd/(4t)$.
- Thin sphere: $\sigma = pd/(4t)$ in all directions.
- Volumetric strain of cylinder: $(pd/4tE)(5 - 4\nu)$; sphere: $(3pd/4tE)(1 - \nu)$. Joint efficiency η: $\sigma_h = pd/(2t\eta_l)$, $\sigma_l = pd/(4t\eta_c)$.

- Thick cylinders (Lamé): $\sigma_r = A - B/r^2$, $\sigma_h = A + B/r^2$; max hoop stress at inner radius.

Shear force and bending moment diagrams

- Relations: $dV/dx = -w$ (load intensity; sign conventions vary), $dM/dx = V$. Max bending moment where SF = 0 (or changes sign). Point load → step in SFD; UDL → linear SFD and parabolic BMD; uniformly varying load → parabolic SFD and cubic BMD. Applied couple → jump in BMD.
- Point of contraflexure: BM changes sign (only in overhanging/fixed beams).
- Standard maxima: simply supported, central load P: $M_{max} = PL/4$; UDL w: $wL^2/8$. Cantilever end load P: PL; UDL: $wL^2/2$. Fixed–fixed UDL: $wL^2/12$ at supports, $wL^2/24$ at centre.

Bending and shear stresses in beams

- Flexure formula: $M/I = \sigma/y = E/R$; $\sigma_{max} = M/Z$, section modulus $Z = I/y_{max}$. Neutral axis passes through the centroid (zero bending stress).
- I values: rectangle $bh^3/12$ ($Z = bh^2/6$); circle $\pi d^4/64$ ($Z = \pi d^3/32$); hollow circle $\pi(D^4 - d^4)/64$; triangle about base $bh^3/12$, about centroid $bh^3/36$.
- Shear stress $\tau = VQ/(Ib)$ (Q = first moment of area above the point). Rectangle: $\tau_{max} = 1.5\tau_{avg}$ at neutral axis (parabolic distribution). Circle: $\tau_{max} = (4/3)\tau_{avg}$. I-section: web carries most shear.
- Beam of uniform strength: section varies so that M/Z is constant.

Shear centre

- Point through which a transverse load must act to cause bending without twisting. Coincides with the centroid for doubly symmetric sections; lies on the axis of symmetry for singly symmetric sections.
- Channel section: shear centre lies outside the web, at $e = b^2h^2t/(4I)$ from the web centre line (b = flange width, h = web height between flange centres, t = flange thickness). Angle and T sections: at the intersection of the legs/flange-web centre lines.

Deflection of beams

- $EI\, d^2y/dx^2 = M$. Methods: double integration, Macaulay's method, moment-area theorems (slope change = area of M/EI diagram; deflection = moment of that area), conjugate beam, superposition, energy methods.

Case	Max slope	Max deflection
Cantilever, end load P	$PL^2/(2EI)$	$PL^3/(3EI)$
Cantilever, UDL w	$wL^3/(6EI)$	$wL^4/(8EI)$
Cantilever, end moment M	$ML/(EI)$	$ML^2/(2EI)$
Simply supported, central P	$PL^2/(16EI)$	$PL^3/(48EI)$
Simply supported, UDL w	$wL^3/(24EI)$	$5wL^4/(384EI)$
Fixed–fixed, central P	0	$PL^3/(192EI)$
Fixed–fixed, UDL w	0	$wL^4/(384EI)$

Torsion of circular shafts

- $T/J = \tau/r = G\theta/L$. Polar moment $J = \pi d^4/32$ (solid), $\pi(D^4 - d^4)/32$ (hollow). $\tau_{max} = 16T/(\pi d^3)$ (solid).
- Angle of twist $\theta = TL/(GJ)$; torsional stiffness GJ/L. Power $P = 2\pi NT/60$ (N in rpm).
- Series shafts: same torque, angles add. Parallel (both ends fixed, torque at a point): same angle, torques add.

- Hollow shafts are stronger and stiffer per unit weight. Combined bending and torsion: equivalent torque $T_e = \sqrt{(M^2 + T^2)}$; equivalent moment $M_e = ½[M + \sqrt{(M^2 + T^2)}]$.
- Closely coiled helical spring: $\tau = 8WDK/(\pi d^3)$ (K = Wahl factor); deflection $\delta = 8WD^3n/(Gd^4)$; stiffness $k = Gd^4/(8D^3n)$. Springs in series: $1/k = \Sigma 1/k_i$; parallel: $k = \Sigma k_i$.

Euler's theory of columns

- Critical buckling load $P_{cr} = \pi^2 EI_{min}/L_e^2$.
- Effective length L_e: both ends pinned L; one fixed one free 2L; both fixed L/2; one fixed one pinned $L/\sqrt{2}$.
- Slenderness ratio $\lambda = L_e/k$ ($k = \sqrt{(I/A)}$, radius of gyration); critical stress $\sigma_{cr} = \pi^2 E/\lambda^2$. Euler valid for long columns (λ above the limit where $\sigma_{cr} = \sigma_{yield}$; ≈ 80–100 for mild steel).
- Rankine–Gordon (short and intermediate columns): $1/P_R = 1/P_{crushing} + 1/P_{Euler}$.
- Eccentric loading: secant formula. Buckling occurs about the axis of minimum I.

Castigliano's theorems (energy methods)

- Strain energy: axial $\int N^2dx/(2AE)$; bending $\int M^2dx/(2EI)$; torsion $\int T^2dx/(2GJ)$; shear $\int kV^2dx/(2GA)$.
- **Castigliano's (second) theorem:** deflection at the point of load P_i in its direction $\delta_i = \partial U/\partial P_i$; rotation $\theta = \partial U/\partial M$. (First theorem: $P_i = \partial U/\partial \delta_i$.) To find deflection where no load acts, apply a dummy load Q and set Q = 0 after differentiating.
- Statically indeterminate structures: $\partial U/\partial R = 0$ for a redundant reaction R (theorem of least work).
- Maxwell's reciprocal theorem: $\delta_{ij} = \delta_{ji}$.

Thermal stresses

- Free thermal strain $\varepsilon_T = \alpha \Delta T$; free expansion $\delta = \alpha L \Delta T$ (no stress).
- Fully restrained: $\sigma = E\alpha\Delta T$ (compressive when heated). Partially restrained with gap δ_0: $\sigma = E(\alpha L\Delta T - \delta_0)/L$.
- Composite bars of different materials heated together: compatibility of strain; the one with higher α goes into compression. $\sigma_1 A_1 = \sigma_2 A_2$ (equilibrium); $\alpha_1\Delta T - \sigma_1/E_1 = \alpha_2\Delta T + \sigma_2/E_2$.

Strain gauges and rosettes

- Strain gauge: $GF = (\Delta R/R)/\varepsilon$. Measures normal strain in one direction only; shear strain must be derived.
- **Rosette** (three gauges at known angles) gives ε_x, ε_y, γ_{xy} using $\varepsilon_\theta = \varepsilon_x \cos^2\theta + \varepsilon_y \sin^2\theta + \gamma_{xy} \sin\theta \cos\theta$.
- Rectangular (0°, 45°, 90°): $\varepsilon_x = \varepsilon_0$, $\varepsilon_y = \varepsilon_{90}$, $\gamma_{xy} = 2\varepsilon_{45} - \varepsilon_0 - \varepsilon_{90}$. Delta (0°, 60°, 120°) rosettes also used.
- Principal stresses from principal strains: $\sigma_1 = E(\varepsilon_1 + \nu\varepsilon_2)/(1 - \nu^2)$; $\sigma_2 = E(\varepsilon_2 + \nu\varepsilon_1)/(1 - \nu^2)$.

Testing of materials

- **Universal testing machine (UTM):** tension, compression, bending, shear tests. Results: E, yield strength (0.2% offset proof stress for materials without a clear yield point), UTS, % elongation = $(L_f - L_0)/L_0 \times 100$, % reduction in area (ductility). Engineering vs true stress ($\sigma_{true} = \sigma(1 + \varepsilon)$, $\varepsilon_{true} = \ln(1 + \varepsilon)$). Necking starts at UTS.
- **Hardness:** Brinell (hardened steel/tungsten ball, $BHN = 2P/[\pi D(D - \sqrt{(D^2 - d^2)})]$, soft-to-medium metals); Vickers (diamond pyramid 136°, $HV = 1.854P/d^2$, all hardness ranges, micro-hardness); Rockwell (depth of penetration, minor + major load; scales B (ball, soft) and

C (diamond cone, hard)); Knoop (thin layers); Shore scleroscope (rebound) and Mohs (scratch). UTS ≈ 3.45 × BHN (MPa) for steels.

- **Impact strength:** Charpy (simply supported, notch facing away from striker) and Izod (cantilever, notch facing striker). Energy absorbed = $W(h_1 - h_2)$. Ductile-to-brittle transition temperature (BCC metals).
- Others: fatigue (rotating beam), creep, torsion test, non-destructive testing (ultrasonic, radiography, dye penetrant, magnetic particle, eddy current).

B2.2 Kinematics and Dynamics of Machines

Mechanisms basics

- Kinematic link, pair (lower: surface contact; higher: point/line contact), chain, mechanism (one link fixed), machine. Inversion: fixing different links of the same chain.
- Mobility (Grübler, planar): $F = 3(n - 1) - 2j - h$ (j = lower pairs, h = higher pairs). $F = 0$ structure; $F < 0$ indeterminate; $F = 1$ constrained mechanism.
- **Grashof's law** (four-bar): $s + l \le p + q \Rightarrow$ at least one link can fully rotate. Shortest link as crank → crank-rocker; shortest as frame → double crank (drag link); shortest as coupler → double rocker. Non-Grashof → triple rocker.
- Inversions: four-bar (beam engine, coupling rod of locomotive, Watt's indicator); single slider-crank (reciprocating engine, Whitworth quick return, rotary engine, oscillating cylinder, hand pump); double slider-crank (elliptical trammel, Scotch yoke, Oldham coupling).
- Quick-return ratio = time of cutting / time of return = α/β (crank angles). Transmission angle μ (best near 90°, acceptable 40°–140°). Toggle position: mechanical advantage → ∞.

Displacement, velocity and acceleration analysis

- **Relative velocity:** $v_{BA} = \omega \cdot AB$, perpendicular to AB. Velocity polygons drawn link by link.
- **Instantaneous centres:** number of ICs $N = n(n - 1)/2$. **Kennedy's theorem:** the three ICs of three bodies in plane motion lie on a straight line. Velocity v = ω × distance from IC. Slider on a fixed straight path: IC at infinity perpendicular to the path.
- **Acceleration:** centripetal (radial) $a_r = \omega^2 r = v^2/r$ towards the centre; tangential $a_t = \alpha r$. **Coriolis component** $2\omega v$ (when a point slides along a rotating link; direction: rotate the sliding velocity by 90° in the sense of ω) — appears in quick-return and slotted-link mechanisms.
- **Slider-crank** (crank r, connecting rod l, $n = l/r$): piston displacement $x \approx r[(1 - \cos\theta) + (\sin^2\theta)/(2n)]$; velocity $v \approx \omega r(\sin\theta + \sin 2\theta/(2n))$; acceleration $a \approx \omega^2 r(\cos\theta + \cos 2\theta/n)$. Angular velocity of connecting rod $\omega_c \approx \omega \cos\theta/n$.
- Analytical: loop-closure equations; graphical: polygons; Klein's construction for slider-crank.

Dynamic analysis of linkages

- D'Alembert: add inertia force $-m\,a_G$ and inertia couple $-I_G\alpha$; then solve as a static problem (link by link, or by virtual work: $\Sigma(F \cdot v) + \Sigma(T \cdot \omega) = 0$).
- Slider-crank engine: piston effort F_P = gas force − inertia force ($F_I = m_R\omega^2 r(\cos\theta + \cos 2\theta/n)$); force along connecting rod $F_Q = F_P/\cos\varphi$; crank effort/torque $T = F_P \cdot r \cdot \sin(\theta + \varphi)/\cos\varphi \approx F_P r(\sin\theta + \sin 2\theta/(2\sqrt{(n^2 - \sin^2\theta)}))$.
- Dynamically equivalent two-mass system: same total mass, same centroid, same moment of inertia ($l_1 l_2 = k^2$).
- **Turning moment diagram and flywheel:** fluctuation of energy $\Delta E = I\omega^2 C_s = mk^2\omega^2 C_s$; coefficient of fluctuation of speed $C_s = (\omega_{max} - \omega_{min})/\omega_{mean}$. Rim flywheel: $\Delta E = mv^2 C_s$.

- Governors (Watt, Porter, Proell, Hartnell): control mean speed with load; sensitiveness, stability, isochronism, hunting. Flywheel controls cyclic fluctuations within a cycle.

Gears and gear trains

- **Law of gearing:** the common normal at the point of contact passes through a fixed point (pitch point) → constant velocity ratio. Satisfied by involute and cycloidal profiles. Involute: centre-distance changes do not affect the velocity ratio; pressure angle constant (14.5°, 20°).
- Terminology: module m = d/T (mm); circular pitch p = πm; diametral pitch P = T/d; addendum = 1m (standard), dedendum = 1.25m; base circle $d_b = d \cos \varphi$. Velocity ratio $\omega_1/\omega_2 = T_2/T_1 = d_2/d_1$.
- Path of contact = $\sqrt{(R_a^2 - R^2\cos^2\varphi)} + \sqrt{(r_a^2 - r^2\cos^2\varphi)} - (R + r)\sin\varphi$. Arc of contact = path/cos φ. **Contact ratio** = arc of contact/circular pitch (> 1, usually 1.2–1.6).
- **Interference:** tip of one gear digs into the non-involute flank of the other. Minimum teeth on pinion to avoid interference (rack, 20° full depth): $T_{min} = 2a_w/\sin^2\varphi \approx 17$ (32 for 14.5°). Remedies: increase pressure angle, stub teeth, undercutting, profile shift.
- Gear types: spur, helical (axial thrust; double helical/herringbone cancels it), bevel (intersecting shafts), worm and wheel (large ratio, non-intersecting perpendicular shafts, often self-locking), rack and pinion, hypoid, spiral.
- **Gear trains:** simple (idler only changes direction), compound (speed ratio = product of driven teeth/product of driver teeth), reverted (input and output coaxial).
- **Epicyclic (planetary)** trains: arm rotates; use the tabular method (lock arm, give +1 rotation to sun, then add arm rotation) or the relative velocity formula $(N_s - N_a)/(N_r - N_a) = -T_r/T_s$. Ring teeth $T_r = T_s + 2T_p$ (same module). Torque balance: $\Sigma T = 0$ ($T_{in} + T_{out} + T_{fixed} = 0$) and power balance $\Sigma T\omega = 0$. Used in automatic transmissions, differentials, robot reducers.
- Harmonic drives (strain wave): very high ratio, zero backlash — common in robot joints.

Balancing of rotating masses

- Unbalanced force from mass m at radius r: $F = m\omega^2 r$.
- **Static balance:** $\Sigma mr = 0$ (vector sum; one balancing mass in the same plane). **Dynamic balance:** $\Sigma mr = 0$ and $\Sigma mrl = 0$ (couple about a reference plane); requires two balancing masses in two planes.
- Graphical: force polygon and couple polygon (reference plane chosen at one balancing plane to eliminate its unknown).

Balancing of reciprocating masses

- Primary disturbing force $m\,\omega^2 r \cos\theta$; secondary $m\,\omega^2 r \cos 2\theta/n$ (behaves like a mass at radius r/(4n) rotating at 2ω).
- Single cylinder: primary force can be only partially balanced (fraction c of the reciprocating mass); adding rotating balance mass introduces an unbalanced vertical force $c\,m\omega^2 r \sin\theta$. Typically c = 2/3 to 3/4 in locomotives (hammer blow, swaying couple, variation of tractive effort).
- Multi-cylinder in-line: Σ primary forces and couples = 0, Σ secondary forces and couples = 0 (use direct and reverse crank method for V-engines and radial engines). In-line 6-cylinder engines are fully balanced for primary and secondary.

Gyroscope

- Gyroscopic couple $C = I\omega\omega_p$ (I = spin moment of inertia, ω = spin, ω_p = precession). The spin vector moves towards the applied torque vector (precession).

- Ship: steering (turning) → pitching effect on bow/stern; pitching → yawing (steering) effect; rolling → no gyroscopic effect (spin and roll axes parallel). Aeroplane turning → nose up or down depending on directions.
- Vehicles on curved track: gyroscopic couple of wheels and engine changes wheel reactions, adds to overturning effect. Two-wheeler cornering requires leaning.
- Gyroscopes in navigation/stabilization (and MEMS gyros in robot IMUs).

Vibrations of single DOF systems

- **Free undamped:** $m\ddot{x} + kx = 0$; $\omega_n = \sqrt{(k/m)}$; $f_n = \omega_n/2\pi$; static deflection $\delta_{st} = mg/k \rightarrow \omega_n = \sqrt{(g/\delta_{st})}$. Energy method (Rayleigh): $d(KE + PE)/dt = 0$.
- Examples: simple pendulum $\omega_n = \sqrt{(g/L)}$; compound pendulum $\omega_n = \sqrt{(mgh/I_O)}$; torsional $\omega_n = \sqrt{(k_t/I)}$; spring mass with spring mass m_s: add $m_s/3$. Cantilever end mass $k = 3EI/L^3$; simply supported centre $k = 48EI/L^3$.
- **Free damped:** $m\ddot{x} + c\dot{x} + kx = 0$. Critical damping $c_c = 2\sqrt{(km)} = 2m\omega_n$; damping ratio $\zeta = c/c_c$. $\zeta < 1$ underdamped: $\omega_d = \omega_n\sqrt{(1 - \zeta^2)}$. Logarithmic decrement $\delta = \ln(x_n/x_{n+1}) = 2\pi\zeta/\sqrt{(1 - \zeta^2)} \approx 2\pi\zeta$; over n cycles $\delta = (1/n)\ln(x_0/x_n)$.
- Coulomb (dry friction) damping: amplitude decreases linearly by $4F/k$ per cycle.
- **Forced vibration** $F_0 \sin \omega t$: steady-state amplitude $X = (F_0/k)/\sqrt{[(1 - r^2)^2 + (2\zeta r)^2]}$, $r = \omega/\omega_n$. Magnification factor $MF = X/(F_0/k)$. Phase $\tan \varphi = 2\zeta r/(1 - r^2)$ (90° at resonance).
- Peak amplitude at $r = \sqrt{(1 - 2\zeta^2)}$; at resonance ($r = 1$) $MF = 1/(2\zeta)$. For $r \gg 1$ the mass barely moves ($X \rightarrow F_0/(m\omega^2)$).
- Rotating unbalance m_0e: $X = (m_0e/m)\cdot r^2/\sqrt{[(1 - r^2)^2 + (2\zeta r)^2]}$. Base excitation: same as transmissibility below.

Effect of damping; vibration isolation

- Damping reduces amplitude near resonance strongly, has little effect far from resonance.
- **Transmissibility** $TR = F_T/F_0 = \sqrt{[1 + (2\zeta r)^2]}/\sqrt{[(1 - r^2)^2 + (2\zeta r)^2]}$.
- $TR = 1$ at $r = \sqrt{2}$ for any ζ. Isolation ($TR < 1$) only when $r > \sqrt{2}$ → use soft mounts (low k, low ω_n). In the isolation region, more damping increases TR (worse), but damping helps when passing through resonance during start-up.
- Isolation efficiency = 1 − TR.

Resonance and critical speed of shafts

- Resonance: forcing frequency = natural frequency → large amplitudes, limited only by damping.
- Critical (whirling) speed: rotor speed at which the shaft whirls violently = its natural transverse frequency, $\omega_c = \sqrt{(k/m)} = \sqrt{(g/\delta_{st})}$; $N_c = (60/2\pi)\sqrt{(g/\delta)}$. Shaft with eccentricity e: whirl amplitude $y = e\, r^2/(1 - r^2)$ (undamped); above critical speed the shaft centre tends towards the mass centre (self-centring).
- Multiple loads — Dunkerley: $1/\omega_c^2 = 1/\omega_1^2 + 1/\omega_2^2 + \ldots + 1/\omega_s^2$ (lower bound). Rayleigh's method for the fundamental frequency (upper bound).
- Two-rotor torsional system: node location from $I_A l_A = I_B l_B$; $\omega = \sqrt{(GJ/(I_A l_A))}$.

B2.3 Machine Design and Computer Integrated Manufacturing

Design for static loading

- Allowable stress = σ_y/FoS (ductile) or σ_u/FoS (brittle). Factor of safety accounts for uncertainty in load, material, analysis.

- **Stress concentration** $K_t = \sigma_{max}/\sigma_{nom}$ (holes, notches, fillets, keyways). Elliptical hole in a plate: $K_t = 1 + 2a/b$; circular hole: 3. Important for brittle materials under static load and for all materials under fatigue; reduce with generous fillets, relief grooves.
- Combined loads: superpose normal stresses; compute principal stresses; apply a failure theory.

Failure theories

Theory	Criterion	Use
Max principal stress (Rankine)	$\sigma_1 = S_{ut}/N$	Brittle materials
Max shear stress (Tresca/Guest)	$\sigma_1 - \sigma_3 = S_y/N$; $\tau_{max} = S_y/(2N)$; shear yield $S_{sy} = 0.5S_y$	Ductile, conservative; hexagon in σ_1–σ_2 plane
Distortion energy (von Mises)	$\sigma_{vm} = \sqrt{(\sigma_1^2 - \sigma_1\sigma_2 + \sigma_2^2)} = S_y/N$; general $\sqrt{[\tfrac{1}{2}((\sigma_1-\sigma_2)^2 + (\sigma_2-\sigma_3)^2 + (\sigma_3-\sigma_1)^2)]}$; $S_{sy} = 0.577S_y$	Ductile, most accurate; ellipse
Max principal strain (St. Venant)	$\sigma_1 - \nu(\sigma_2 + \sigma_3) = S_y/N$	Rarely used
Max strain energy (Haigh)	$\sigma_1^2 + \sigma_2^2 - 2\nu\sigma_1\sigma_2 = (S_y/N)^2$	Ductile (less used)
Coulomb–Mohr	$\sigma_1/S_t - \sigma_3/S_c = 1/N$	Brittle with $S_c \neq S_t$

- Shaft under bending + torsion (Tresca): $\tau_{max} = (16/\pi d^3)\sqrt{(M^2 + T^2)}$; (von Mises): $\sigma_{eq} = (32/\pi d^3)\sqrt{(M^2 + 0.75T^2)}$.

Fatigue strength and the S-N diagram

- Fatigue: failure under fluctuating stress well below S_u; crack initiation at stress raisers, propagation (beach marks), sudden final fracture.
- **S-N (Wöhler) diagram:** log stress amplitude vs log cycles. Steels show an endurance limit $S_e' \approx 0.5S_{ut}$ (for $S_{ut} \leq 1400$ MPa) at ~10^6–10^7 cycles; non-ferrous metals (aluminium) have no endurance limit. Low-cycle (<10^3) vs high-cycle fatigue; 10^3 cycles $\approx 0.9S_{ut}$.
- Corrected endurance limit (Marin): $S_e = k_a k_b k_c k_d k_e S_e'$ (surface, size, load, temperature, reliability factors). Fatigue stress concentration $K_f = 1 + q(K_t - 1)$, notch sensitivity $0 \leq q \leq 1$.
- Fluctuating stress: mean $\sigma_m = (\sigma_{max} + \sigma_{min})/2$; amplitude $\sigma_a = (\sigma_{max} - \sigma_{min})/2$; stress ratio $R = \sigma_{min}/\sigma_{max}$ (fully reversed $R = -1$).
- Mean-stress criteria: **Soderberg** $\sigma_a/S_e + \sigma_m/S_y = 1/N$ (most conservative); **Goodman** $\sigma_a/S_e + \sigma_m/S_{ut} = 1/N$; **Gerber** parabola $N\sigma_a/S_e + (N\sigma_m/S_{ut})^2 = 1$.
- Cumulative damage — Miner's rule: $\Sigma n_i/N_i = 1$ at failure.
- Improve fatigue life: polish surfaces, shot peening (compressive residual stress), avoid sharp corners, nitriding/case hardening.

Bolted joints

- Tensile stress area; $\sigma = P/A_t$. Core diameter $d_c \approx 0.84d$. Shear of bolts in single/double shear: $P = n\cdot(\pi/4)d^2\cdot\tau$ (×2 for double).
- Preload (initial tension) F_i; external load P shared: bolt load $F_b = F_i + C\cdot P$, $C = k_b/(k_b + k_m)$ (joint stiffness constant); joint separates when $(1 - C)P = F_i$. Tightening torque $T \approx 0.2F_i d$.

- **Eccentrically loaded bolted joints:** primary (direct) shear P/n on each bolt + secondary shear from moment ∝ distance from the centroid of the bolt group: $F''_i = P \cdot e \cdot r_i / \Sigma r^2$; combine vectorially. For a moment in the plane perpendicular to bolts (bracket), tensile load ∝ distance from the tilting edge: $F_i = P \cdot e \cdot l_i / \Sigma l^2$.
- Power screws: torque to raise $T = W(d_m/2)\tan(\varphi + \alpha)$ (square thread); self-locking if $\varphi > \alpha$; efficiency $\eta = \tan\alpha / \tan(\alpha + \varphi) < 50\%$ if self-locking.

Riveted joints

- Lap and butt joints (single/double strap). Failure modes per pitch length: tearing of plate $P_t = (p - d)t\sigma_t$; shearing of rivets $P_s = n(\pi/4)d^2\tau$ (×2 for double shear; IBR: ×1.875); crushing $P_c = n \cdot d \cdot t \cdot \sigma_c$.
- Efficiency $\eta = \min(P_t, P_s, P_c)/(p \cdot t \cdot \sigma_t)$. Unwin's formula $d = 6\sqrt{t}$ (mm). Caulking and fullering for leak-proof boiler joints.

Welded joints

- Butt weld: $P = \sigma_t \cdot t \cdot l$. Fillet weld throat $t = 0.707s$ (leg size s). Transverse fillet: $P = 0.707s \cdot l \cdot \sigma_t$; parallel (longitudinal) fillet: $P = 0.707s \cdot l \cdot \tau$ (failure in shear along the throat).
- Eccentric loading: primary shear P/A plus secondary shear $\tau_2 = M \cdot r / J$ (J = polar moment of the weld throat area, using unit-weld properties $J_u \times 0.707s$); combine vectorially. Bending: $\sigma_b = M/Z$ of the weld.
- Stress concentration factors for welds in fatigue (reinforced butt weld 1.2, toe of transverse fillet 1.5, end of parallel fillet 2.7).

Shafts, keys and couplings

- Shaft design by strength: ASME code $d^3 = (16/\pi\tau_{allow})\sqrt{[(K_b M)^2 + (K_t T)^2]}$ (shock/fatigue factors). By rigidity: torsional $\theta = TL/(GJ)$ limit (e.g. 0.25°/m) and lateral deflection limits.
- Hollow vs solid: same weight hollow shaft is stronger and stiffer.
- Keys: sunk (rectangular/square), saddle, feather, Woodruff, splines. Square key checked for shear ($\tau = 2T/(d \cdot w \cdot l)$) and crushing ($\sigma_c = 4T/(d \cdot h \cdot l)$); square key equally strong in both when $\sigma_c = 2\tau$.
- Couplings: rigid (sleeve, flange) and flexible (bushed-pin, Oldham, universal/Hooke's joint). Hooke's joint: speed ratio $\omega_2/\omega_1 = \cos\alpha / (1 - \cos^2\theta \sin^2\alpha)$; varies between $\cos\alpha$ and $1/\cos\alpha$; double Hooke's joint gives constant velocity if arranged correctly.

Rolling contact bearings

- Types: deep-groove ball (radial + some axial), angular contact, cylindrical roller (high radial), tapered roller (combined), spherical roller (self-aligning), thrust bearings, needle.
- **Life:** $L_{10} = (C/P)^k$ million revolutions (k = 3 ball, 10/3 roller); C = basic dynamic load rating (90% reliability). $L_h = L_{10} \times 10^6/(60N)$ hours.
- Equivalent dynamic load $P = XVF_r + YF_a$ (V = 1 inner race rotating). Reliability adjustment via Weibull: $L/L_{10} = [\ln(1/R)/\ln(1/0.9)]^{1/b}$, b ≈ 1.17 (bearing-life Weibull slope; 1.5 in some texts).
- Variable load: $P_e = [\Sigma(P_i^k n_i)/\Sigma n_i]^{1/k}$. Static load rating C_0 for stationary/slow bearings.

Sliding contact (journal) bearings

- Hydrodynamic lubrication: wedge-shaped oil film builds pressure due to rotation (Petroff, Reynolds equation). Lubrication regimes: boundary, mixed, hydrodynamic (Stribeck curve: friction vs ZN/p).

- **Sommerfeld number** $S = (r/c)^2 \cdot \mu N/p$ ($p = W/(l \cdot d)$, N in rev/s); design charts (Raimondi–Boyd) give friction coefficient, minimum film thickness, flow, temperature rise. Bearing characteristic number ZN/p must exceed the minimum (≈ 3× critical).
- Petroff's equation: $f = 2\pi^2(\mu N/p)(r/c)$. Hydrostatic bearings: external pump supplies pressure (work at zero speed).
- Bearing materials: babbitt (white metal), bronze, aluminium alloys, PTFE, porous self-lubricating.

Brakes

- **Block (shoe) brake:** $F_{braking} = \mu R_n$; torque $T = \mu R_n r$. Self-energizing if friction moment assists the applied force; self-locking if the required actuating force ≤ 0. Long shoe: equivalent $\mu' = 4\mu \sin\theta/(2\theta + \sin 2\theta)$.
- **Band brake:** $T_1/T_2 = e^{\mu\theta}$; torque $(T_1 - T_2)r$. Simple, differential (can be self-locking) and band-and-block types.
- **Disc brakes, internal expanding shoe brakes** (automobiles). Energy absorbed: KE + PE; temperature rise $\Delta T = E/(m \cdot c)$.
- Dynamometers (absorption: Prony brake, rope brake; transmission types) measure power.

Clutches

- Friction plate clutch torque $T = n\mu W R_m$ (n = number of friction surfaces).
- **Uniform pressure** (new clutch): $R_m = (2/3)(R_o^3 - R_i^3)/(R_o^2 - R_i^2)$. **Uniform wear** (worn clutch, pr = constant): $R_m = (R_o + R_i)/2$. Uniform wear gives lower torque → safer design assumption. Optimum $R_i/R_o \approx 0.577$ (uniform wear).
- Cone clutch: $T = \mu W R_m/\sin\alpha$. Centrifugal clutch: $(F_c - F_s)\mu\, r\, n$. Multi-plate clutches for compact, high torque.

Springs

- Helical compression spring: spring index $C = D/d$ (4–12); Wahl factor $K = (4C - 1)/(4C - 4) + 0.615/C$ (curvature + direct shear); $\tau = K \cdot 8PD/(\pi d^3)$; deflection $\delta = 8PD^3N/(Gd^4)$; stiffness $k = Gd^4/(8D^3N)$.
- Energy stored $U = \tfrac{1}{2}P\delta$. Solid length $= N_t \cdot d$; free length = solid length + max deflection + clearance. Buckling if free length/D is too large. Surge: natural frequency should exceed operating frequency (~15–20×).
- Leaf (laminated) springs: treat each half as a cantilever of length L carrying W: $\sigma = 6WL/(nbt^2)$, $\delta = 6WL^3/(nEbt^3)$ (n leaves, width b, thickness t); nipping (pre-stressing) equalizes stresses in full-length and graduated leaves. Torsion springs, Belleville (disc) springs, spiral springs.

CAD / CAM and integration

- **CAD:** geometric modelling — wireframe (edges only, ambiguous), surface (Bezier, B-spline, NURBS; Bezier curves: degree = control points − 1, global control; B-splines: local control), solid (B-rep and CSG). Parametric and feature-based modelling with constraints; assemblies; drafting; FEA (analysis) and CAE.
- Transformations: homogeneous 2D/3D matrices for translation, rotation, scaling, reflection, shear; composite = matrix product (order matters). Projections: orthographic, isometric, perspective.
- Curve representations: explicit, implicit, parametric (preferred). Hermite cubic: needs end points and tangents. Continuity C^0, C^1, C^2 and G^1.
- **CAM:** generating tool paths and NC code from CAD geometry; post-processors convert to machine-specific G-code. CAPP, NC verification/simulation, robot offline programming.

- **Integration tools/standards:** data exchange formats IGES, STEP (ISO 10303, product data), STL (triangulated surface for 3D printing), DXF (2D), Parasolid/ACIS kernels. PDM/PLM systems manage product data and revisions. CAD/CAM integration → CIM with MRP/ERP, FMS, DNC.
- Group technology (part families, Opitz/MICLASS coding), FMS (flexible manufacturing systems: workstations + automated handling + central computer).

Additive manufacturing (3D printing)

- Workflow: CAD model → STL (tessellation) → orientation and support generation → slicing into layers → layer-by-layer build → post-processing (support removal, curing, sintering, machining).
- ASTM/ISO 52900 categories:
  - **Material extrusion** — FDM/FFF (thermoplastic filament: PLA, ABS, nylon).
  - **Vat photopolymerization** — SLA (laser), DLP (projector); liquid resin cured by UV; high resolution.
  - **Powder bed fusion** — SLS (polymers), SLM/DMLS (metals, laser), EBM (electron beam, vacuum).
  - **Material jetting** — PolyJet (multi-material, colour).
  - **Binder jetting** — binder into powder (sand moulds, metals, then sintered).
  - **Directed energy deposition** — LENS, wire-arc AM (repair, large metal parts).
  - **Sheet lamination** — LOM, ultrasonic AM.
- Advantages: complex geometries (lattices, internal channels), no tooling, customization, rapid prototyping, part consolidation. Limitations: slow for mass production, anisotropy, surface finish (stair-stepping proportional to layer thickness), size limits, support structures.
- Applications: prototypes, jigs and fixtures, robot grippers and custom end-effectors, medical implants, aerospace parts, tooling (conformal cooling channels).